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## Reconfigurable upper leg support for a surgical frame

### Abstract

A surgical frame and method for use thereof is provided. The surgical frame is capable of reconfiguration before, during, or after surgery. The surgical frame includes a main beam that can be rotated, raised/lowered, and tilted upwardly/downwardly to afford positioning and repositioning of a patient supported thereon. The surgical frame also includes a reconfigurable upper leg support for supporting portions of the upper legs, the hips, and the lower back of the patient to facilitate positioning and repositioning there during surgery. The upper leg support via reconfiguration thereof can accommodate patients of different sizes, can provide flexure of the patient's lumbar spine to facilitate surgical access thereto, and can prevent unwanted torsion of a patient's spine during such reconfiguration.

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## Background/Summary

(1) The present application is a continuation of U.S. Ser. No. 16/927,219, filed Jul. 13, 2020; which claims benefit of U.S. Provisional Application No. 62/905,770, filed Sep. 25, 2019; all of which are incorporated by reference herein.

FIELD
(1) The present technology generally relates to a reconfigurable upper leg support for use with a surgical frame incorporating a main beam capable of rotation.
BACKGROUND
(2) Access to a patient is of paramount concern during surgery. Surgical frames have been used to position and reposition patients during surgery. For example, surgical frames have been configured to manipulate the rotational position of the patient before, during, and even after surgery. Such surgical frames include support structures to facilitate the rotational movement of the patient. Typical support structures can include main beams supported at either end thereof for rotational movement about axes of rotation extending along the lengths of the surgical frames. The main beams can be positioned and repositioned to afford various positions of the patients positioned thereon. To illustrate, the main beams can be rotated for positioning a patient in prone positions, lateral positions, and positions 45° between the prone and lateral positions. In addition to the rotational positioning afforded by the main beams, the patients can be further manipulated by support structures attached relative to the main beam. To illustrate, an upper leg support can be provided to support portions of upper legs, hips, and the lower back of the patient. Such an upper leg support can be moveable with respect to the main beam to facilitate positioning and repositioning of the upper legs, the hips, and the lower back of the patient to facilitate access to the patient during surgery. However, patients have different sizes and it is desirous to inhibit torsion of the patient's spine during use of surgical frame. Therefore, there is a need for a reconfigurable upper leg support that via reconfiguration thereof can accommodate patients of different sizes, can provide flexure of the patient's lumbar spine to facilitate surgical access thereto, and can prevent unwanted torsion of a patient's spine during such reconfiguration.
SUMMARY
(3) The techniques of this disclosure generally relate to a reconfigurable upper leg support attached

relative to a rotatable main beam that is articulable to adjust the position of the upper legs of a patient to correspondingly affect the flexure of the lumbar spine of a patient, while simultaneously inhibiting unwanted torsion of the patient's spine caused by reconfiguration of the upper leg support.

(4) In one aspect, the present disclosure provides a method of adjusting a position of a patient supported on a surgical frame, the method including positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate; extending a first arm portion relative to a second arm portion to adjust a position of a platform portion relative to a portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached relative to the platform portion, and the second arm portion including a first end attached relative to the platform portion and a second end attached relative to the portion of the surgical frame; extending a first telescoping shaft to adjust a position of the platform portion relative to at least one of the first arm portion and the second arm portion, the first telescoping shaft including a first end attached relative to the second arm portion and a second end attached relative to the platform portion; extending a second telescoping shaft to adjust a position of the support plate relative to the platform portion, the second telescoping shaft including a first end attached relative to the platform portion and a second end attached to a support bracket moveably attached relative to the support platform, and the support plate being supported relative to the support bracket; adjusting a center of rotation of a lumbar portion of a spine of the patient by coordinating the extension of the first arm portion, the first telescoping shaft, and the second telescoping shaft.

(5) In one aspect, the present disclosure provides a method of adjusting a position of a patient supported on a surgical frame, the method including supporting upper legs of the patient on a support plate; extending a first arm portion to adjust a position of a platform portion relative to a portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached relative to the platform portion; extending a telescoping shaft to adjust a position of the support plate relative to the platform portion, the telescoping shaft including a first end attached relative to the platform portion and a second end attached to a support bracket moveably attached relative to the support platform, and the support plate being supported relative to the support bracket; and adjusting a center of rotation of a lumbar portion of a spine of the patient by coordinating the extension of the first arm portion and the telescoping shaft.

(6) In one aspect, the present disclosure provides a method of adjusting a position of a patient supported on a surgical frame, the method including positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate; pivoting an arm portion relative to a portion of the surgical frame to adjust a position of a platform portion relative to a portion of the surgical frame, the arm portion including a first end pivotally attached relative to the portion of the surgical frame and a second end attached relative to the platform portion; extending a first telescoping shaft to adjust a position of the platform portion relative to the arm portion, the first telescoping shaft including a first end attached relative to the arm portion and a second end attached relative to the platform portion; and extending a second telescoping shaft to adjust a position of the support plate relative to the platform portion, the second telescoping shaft including a first end attached relative to the platform portion and a second end attached to a support bracket moveably attached relative to the support platform, and the support plate being supported relative to the support bracket.

(7) The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

## Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a top perspective view that illustrates a prior art surgical frame with a patient positioned thereon in a prone position;

(2) FIG. **2** is a side elevational view that illustrates the surgical frame of FIG. **1** with the patient positioned thereon in a prone position;

(3) FIG. **3** is another side elevational view that illustrates the surgical frame of FIG. **1** with the patient positioned thereon in a prone position;

(4) FIG. **4** is a top plan view that illustrates the surgical frame of FIG. **1** with the patient positioned thereon in a prone position;

(5) FIG. **5** is a top perspective view that illustrates the surgical frame of FIG. **1** with the patient positioned thereon in a lateral position;

(6) FIG. **6** is a top perspective view that illustrates portions of the surgical frame of FIG. **1** showing an area of access to the head of the patient positioned thereon in a prone position;

(7) FIG. **7** is a side elevational view that illustrates the surgical frame of FIG. **1** showing a torso-lift support supporting the patient in a lifted position;

(8) FIG. **8** is another side elevational view that illustrates the surgical frame of FIG. **1** showing the torso-lift support supporting the patient in the lifted position;

(9) FIG. **9** is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. **1** showing the torso-lift support supporting the patient in an unlifted position;

(10) FIG. **10** is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. **1** showing the torso-lift support supporting the patient in the lifted position;

(11) FIG. **11** is an enlarged top perspective view that illustrates componentry of the torso-lift support in the unlifted position;

(12) FIG. **12** is an enlarged top perspective view that illustrates the componentry of the torso-lift support in the lifted position;

(13) FIG. **13**A is a perspective view of an embodiment that illustrates a structural offset main beam for use with another embodiment of a torso-lift support showing the torso-lift support in a retracted position;

(14) FIG. **13**B is a perspective view similar to FIG. **13**A showing the torso-lift support at half travel;

(15) FIG. **13**C is a perspective view similar to FIGS. **13**A and **13**B showing the torso-lift support at full travel;

(16) FIG. **14** is a perspective view that illustrates a chest support lift mechanism of the torso-lift support of FIGS. **13**A-**13**C with actuators thereof retracted;

(17) FIG. **15** is another perspective view that illustrates a chest support lift mechanism of the torso-lift support of FIGS. **13**A-**13**C with the actuators thereof extended;

(18) FIG. **16** is a top perspective view that illustrates the surgical frame of FIG. **1**;

(19) FIG. **17** is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. **1** showing a sagittal adjustment assembly including a pelvic-tilt mechanism and leg adjustment mechanism;

(20) FIG. **18** is an enlarged side elevational view that illustrates portions of the surgical frame of FIG. **1** showing the pelvic-tilt mechanism;

(21) FIG. **19** is an enlarged perspective view that illustrates componentry of the pelvic-tilt mechanism;

(22) FIG. **20** is an enlarged perspective view that illustrates a captured rack and a worm gear assembly of the componentry of the pelvic-tilt mechanism;

(23) FIG. **21** is an enlarged perspective view that illustrates the worm gear assembly of FIG. **20**;

(24) FIG. **22** is a side elevational view that illustrates portions of the surgical frame of FIG. **1** showing the patient positioned thereon and the pelvic-tilt mechanism of the sagittal adjustment assembly in the flexed position;
(25) FIG. **23** is another side elevational view that illustrates portions of the surgical frame of FIG. **1** showing the patient positioned thereon and the pelvic-tilt mechanism of the sagittal adjustment assembly in the fully extended position;
(26) FIG. **24** is an enlarged top perspective view that illustrates portions of the surgical frame of FIG. **1** showing a coronal adjustment assembly;
(27) FIG. **25** is a top perspective view that illustrates portions of the surgical frame of FIG. **1** showing operation of the coronal adjustment assembly;
(28) FIG. **26** is a top perspective view that illustrates a portion of the surgical frame of FIG. **1** showing operation of the coronal adjustment assembly;
(29) FIG. **27** is a top perspective view that illustrates a prior art surgical frame in accordance with an embodiment of the present invention with the patient positioned thereon in a prone position showing a translating beam thereof in a first position;
(30) FIG. **28** is another top perspective view that illustrates the surgical frame of FIG. **27** with the patient in a prone position showing the translating beam thereof in a second position;
(31) FIG. **29** is yet another top perspective view that illustrates the surgical frame of FIG. **27** with the patient in a lateral position showing the translating beam thereof in a third position;
(32) FIG. **30** is a top plan view that illustrates the surgical frame of FIG. **27** with the patient in a lateral position showing the translating beam thereof in the third position;
(33) FIG. **31** is a top, side perspective view that illustrates a portion of a main beam of a surgical frame, and a portion of a reconfigurable upper leg support of a first embodiment of the present disclosure;
(34) FIG. **32**A is a top, side perspective view similar to FIG. **31** that illustrates a portion of the reconfigurable upper leg support of FIG. **31** relative to the main beam;
(35) FIG. **32**B is a fragmentary, top, side perspective view similar to FIG. **32**A that illustrates a portion of the reconfigurable upper leg support of FIG. **31** relative to the main beam;
(36) FIG. **33**A is a top, side perspective view that illustrates a portion of the main beam, and a portion of the reconfigurable upper leg support of FIG. **31**;
(37) FIG. **33**B is a fragmentary, top, side perspective view similar to FIG. **33**A that illustrates a portion of the reconfigurable upper leg support of FIG. **31** relative to the main beam;
(38) FIG. **34**A is a bottom, side perspective view that illustrates a portion of the main beam, and a portion of the reconfigurable upper leg support of FIG. **31**;
(39) FIG. **34**B is a fragmentary bottom, side perspective view similar to FIG. **34**B that illustrates a portion of the reconfigurable upper leg support of FIG. **31** relative to the main beam;
(40) FIG. **35** is a side elevational view that illustrates the reconfigurable upper leg support of FIG. **31** with a first arm portion, a first telescoping shaft portion, and a second telescoping shaft portion adjusted to position the upper leg support in a first position;
(41) FIG. **36** is a side elevational view that illustrates the reconfigurable upper leg support of FIG. **31** showing a position of upper legs, hips, and lower back of a patient supported thereby with the upper leg support in the first position;
(42) FIG. **37** is a side elevational view that illustrates the reconfigurable upper leg support of FIG. **31** with the first arm portion, the first telescoping shaft portion, and the second telescoping shaft portion adjusted to position the upper leg support in a second position;
(43) FIG. **38** is a side elevational view that illustrates the reconfigurable upper leg support of FIG. **31** showing a position of the upper legs, hips, and lower back of the patient supported thereby with the upper leg support in the second position;
(44) FIG. **39** is a table illustrating extension amounts for the first arm portion, the first telescoping shaft portion, and the second telescoping shaft portion;

(45) FIG. **40** is a side elevational view that illustrates a portion of a main beam of a surgical frame, and a reconfigurable upper leg support of second embodiment of the present disclosure;
(46) FIG. **41** is a top plan view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(47) FIG. **42** is a bottom plan view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(48) FIG. **43** is a side, top perspective view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(49) FIG. **44** is a side, bottom perspective view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(50) FIG. **45** is an enlarged, bottom plan view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(51) FIG. **46** is an enlarged, bottom perspective view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(52) FIG. **47** is another enlarged, bottom perspective view that illustrates the reconfigurable upper leg support of FIG. **40** relative to the main beam;
(53) FIG. **48** is a side elevational view that illustrates the reconfigurable upper leg support of FIG. **40** and a patient partially supported thereby with a first arm portion, a first telescoping shaft portion, and a second telescoping shaft portion adjusted to position the upper leg support and the patient in a first position; and
(54) FIG. **49** is a side elevational view that illustrates the reconfigurable upper leg support of FIG. **40** and the patient partially supported thereby with the first arm portion, the first telescoping shaft portion, and the second telescoping shaft portion adjusted to position the upper leg support and the patient in a second position.
(55) The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

## DETAILED DESCRIPTION

(56) FIGS. **1-26** depict a prior art embodiment and components of a surgical support frame generally indicated by the numeral **10**. FIGS. **1-26** were previously described in U.S. Ser. No. 15/239,256, which is hereby incorporated by reference herein in its entirety. Furthermore, FIGS. **27-30** were previously described in U.S. Ser. No. 15/639,080, which is hereby incorporated by reference herein in its entirety.
(57) As discussed below, the surgical frame **10** serves as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby, and, in doing so, serves to support the patient P such that the patient's spine does not experience unnecessary torsion.
(58) The surgical frame **10** is configured to provide a relatively minimal amount of structure adjacent the patient's spine to facilitate access thereto and to improve the quality of imaging available before and during surgery. Thus, the surgeon's workspace and imaging access are thereby increased. Furthermore, radiolucent or low magnetic susceptibility materials can be used in constructing the structural components adjacent the patient's spine in order to further enhance imaging quality.
(59) The surgical frame **10** has a longitudinal axis and a length therealong. As depicted in FIGS. **1-5**, for example, the surgical frame **10** includes an offset structural main beam **12** and a support structure **14**. The offset main beam **12** is spaced from the ground by the support structure **14**. As discussed below, the offset main beam **12** is used in supporting the patient P on the surgical frame **10** and various support components of the surgical frame **10** that directly contact the patient P (such as a head support **20**, arm supports **22**A and **22**B, torso-lift supports **24** and **160**, a sagittal adjustment assembly **28** including a pelvic-tilt mechanism **30** and a leg adjustment mechanism **32**,

and a coronal adjustment assembly **34**). As discussed below, an operator such as a surgeon can control actuation of the various support components to manipulate the position of the patient's body. Soft straps (not shown) are used with these various support components to secure the patient P to the frame and to enable either manipulation or fixation of the patient P. Reusable soft pads can be used on the load-bearing areas of the various support components.

(60) The offset main beam **12** is used to facilitate rotation of the patient P. The offset main beam **12** can be rotated a full 360° before and during surgery to facilitate various positions of the patient P to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the offset main beam **12** can be positioned to place the patient P in a prone position (e.g., FIGS. **1-4**), a lateral position (e.g., FIG. **5**), and in a position 45° between the prone and lateral positions. Furthermore, the offset main beam **12** can be rotated to afford anterior, posterior, lateral, anterolateral, and posterolateral pathways to the spine. As such, the patient's body can be flipped numerous times before and during surgery without compromising sterility or safety. The various support components of the surgical frame **10** are strategically placed to further manipulate the patient's body into position before and during surgery. Such intraoperative manipulation and positioning of the patient P affords a surgeon significant access to the patient's body. To illustrate, when the offset main beam **12** is rotated to position the patient P in a lateral position, as depicted in FIG. **5**, the head support **20**, the arm supports **22**A and **22**B, the torso-lift support **24**, the sagittal adjustment assembly **28**, and/or the coronal adjustment assembly **34** can be articulated such that the surgical frame **10** is OLIF-capable or DLIF-capable.

(61) As depicted in FIG. **1**, for example, the support structure **14** includes a first support portion **40** and a second support portion **42** interconnected by a cross member **44**. Each of the first and second support portions **40** and **42** include a horizontal portion **46** and a vertical support post **48**. The horizontal portions **46** are connected to the cross member **44**, and casters **50** can be attached to the horizontal portions **46** to facilitate movement of the surgical frame **10**.

(62) The vertical support posts **48** can be adjustable to facilitate expansion and contraction of the heights thereof. Expansion and contraction of the vertical support posts **48** facilitates raising and lowering, respectively, of the offset main beam **12**. As such, the vertical support posts **48** can be adjusted to have equal or different heights. For example, the vertical support posts **48** can be adjusted such that the vertical support post **48** of the second support portion **42** is raised 12 inches higher than the vertical support post **48** of the first support portion **40** to place the patient P in a reverse Trendelenburg position.

(63) Furthermore, cross member **44** can be adjustable to facilitate expansion and contraction of the length thereof. Expansion and contraction of the cross member **44** facilitates lengthening and shortening, respectively, of the distance between the first and second support portions **40** and **42**.

(64) The vertical support post **48** of the first and second support portions **40** and **42** have heights at least affording rotation of the offset main beam **12** and the patient P positioned thereon. Each of the vertical support posts **48** include a clevis **60**, a support block **62** positioned in the clevis **60**, and a pin **64** pinning the clevis **60** to the support block **62**. The support blocks **62** are capable of pivotal movement relative to the clevises **60** to accommodate different heights of the vertical support posts **48**. Furthermore, axles **66** extending outwardly from the offset main beam **12** are received in apertures **68** formed on the support blocks **62**. The axles **66** define an axis of rotation of the offset main beam **12**, and the interaction of the axles **66** with the support blocks **62** facilitate rotation of the offset main beam **12**.

(65) Furthermore, a servomotor **70** can be interconnected with the axle **66** received in the support block **62** of the first support portion **40**. The servomotor **70** can be computer controlled and/or operated by the operator of the surgical frame **10** to facilitate controlled rotation of the offset main beam **12**. Thus, by controlling actuation of the servomotor **70**, the offset main beam **12** and the patient P supported thereon can be rotated to afford the various surgical pathways to the patient's spine.

(66) As depicted in FIGS. **1-5**, for example, the offset main beam **12** includes a forward portion **72** and a rear portion **74**. The forward portion **72** supports the head support **20**, the arm supports **22**A and **22**B, the torso-lift support **24**, and the coronal adjustment assembly **34**, and the rear portion **74** supports the sagittal adjustment assembly **28**. The forward and rear portions **72** and **74** are connected to one another by connection member **76** shared therebetween. The forward portion **72** includes a first portion **80**, a second portion **82**, a third portion **84**, and a fourth portion **86**. The first portion **80** extends transversely to the axis of rotation of the offset main beam **12**, and the second and fourth portions **82** and **86** are aligned with the axis of rotation of the offset main beam **12**. The rear portion **74** includes a first portion **90**, a second portion **92**, and a third portion **94**. The first and third portions **90** and **94** are aligned with the axis of rotation of the offset main beam **12**, and the second portion **92** extends transversely to the axis of rotation of the offset main beam **12**.
(67) The axles **66** are attached to the first portion **80** of the forward portion **72** and to the third portion **94** of the rear portion **74**. The lengths of the first portion **80** of the forward portion **72** and the second portion **92** of the rear portion **74** serve in offsetting portions of the forward and rear portions **72** and **74** from the axis of rotation of the offset main beam **12**. This offset affords positioning of the cranial-caudal axis of patient P approximately aligned with the axis of rotation of the offset main beam **12**.
(68) Programmable settings controlled by a computer controller (not shown) can be used to maintain an ideal patient height for a working position of the surgical frame **10** at a near-constant position through rotation cycles, for example, between the patient positions depicted in FIGS. **1** and **5**. This allows for a variable axis of rotation between the first portion **40** and the second portion **42**.
(69) As depicted in FIG. **5**, for example, the head support **20** is attached to a chest support plate **100** of the torso-lift support **24** to support the head of the patient P. If the torso-lift support **24** is not used, the head support **20** can be directly attached to the forward portion **72** of the offset main beam **12**. As depicted in FIGS. **4** and **6**, for example, the head support **20** further includes a facial support cradle **102**, an axially adjustable head support beam **104**, and a temple support portion **106**. Soft straps (not shown) can be used to secure the patient P to the head support **20**. The facial support cradle **102** includes padding across the forehead and cheeks, and provides open access to the mouth of the patient P. The head support **20** also allows for imaging access to the cervical spine. Adjustment of the head support **20** is possible via adjusting the angle and the length of the head support beam **104** and the temple support portion **106**.
(70) As depicted in FIG. **5**, for example, the arm supports **22**A and **22**B contact the forearms and support the remainder of the arms of the patient P, with the first arm support **22**A and the second arm support **22**B attached to the chest support plate **100** of the torso-lift support **24**. If the torso-lift support **24** is not used, the arm supports **22**A and **22**B can both be directly attached to the offset main beam **12**. The arm supports **22**A and **22**B are positioned such that the arms of the patient P are spaced away from the remainder of the patient's body to provide access (FIG. **6**) to at least portions of the face and neck of the patient P, thereby providing greater access to the patient.
(71) As depicted in FIGS. **7-12**, for example, the surgical frame **10** includes a torso-lift capability for lifting and lowering the torso of the patient P between an uplifted position and a lifted position, which is described in detail below with respect to the torso-lift support **24**. As depicted in FIGS. **7** and **8**, for example, the torso-lift capability has an approximate center of rotation (“COR”) **108** that is located at a position anterior to the patient's spine about the L2 of the lumbar spine, and is capable of elevating the upper body of the patient at least an additional six inches when measured at the chest support plate **100**.
(72) As depicted in FIGS. **9-12**, for example, the torso-lift support **24** includes a “crawling” four-bar mechanism **110** attached to the chest support plate **100**. Soft straps (not shown) can be used to secure the patient P to the chest support plate **100**. The head support **20** and the arm supports **22**A and **22**B are attached to the chest support plate **100**, thereby moving with the chest support plate **100** as the chest support plate **100** is articulated using the torso-lift support **24**. The fixed COR **108**

is defined at the position depicted in FIGS. **7** and **8**. Appropriate placement of the COR **108** is important so that spinal cord integrity is not compromised (i.e., overly compressed or stretched) during the lift maneuver performed by the torso-lift support **24**.

(73) As depicted in FIGS. **10-12**, for example, the four-bar mechanism **110** includes first links **112** pivotally connected between offset main beam **12** and the chest support plate **100**, and second links **114** pivotally connected between the offset main beam **12** and the chest support plate **100**. As depicted in FIGS. **11** and **12**, for example, in order to maintain the COR **108** at the desired fixed position, the first and second links **112** and **114** of the four-bar mechanism **110** crawl toward the first support portion **40** of the support structure **14**, when the patient's upper body is being lifted. The first and second links **112** and **114** are arranged such that neither the surgeon's workspace nor imaging access are compromised while the patient's torso is being lifted.

(74) As depicted in FIGS. **11** and **12**, for example, each of the first links **112** define an L-shape, and includes a first pin **116** at a first end **118** thereof. The first pin **116** extends through first elongated slots **120** defined in the offset main beam **12**, and the first pin **116** connects the first links **112** to a dual rack and pinion mechanism **122** via a drive nut **124** provided within the offset main beam **12**, thus defining a lower pivot point thereof. Each of the first links **112** also includes a second pin **126** positioned proximate the corner of the L-shape. The second pin **126** extends through second elongated slots **128** defined in the offset main beam **12**, and is linked to a carriage **130** of rack and pinion mechanism **122**. Each of the first links **112** also includes a third pin **132** at a second end **134** that is pivotally attached to chest support plate **100**, thus defining an upper pivot point thereof.

(75) As depicted in FIGS. **11** and **12**, for example, each of the second links **114** includes a first pin **140** at a first end **142** thereof. The first pin **140** extends through the first elongated slot **120** defined in the offset main beam **12**, and the first pin **140** connects the second links **114** to the drive nut **124** of the rack and pinion mechanism **122**, thus defining a lower pivot point thereof. Each of the second links **114** also includes a second pin **144** at a second end **146** that is pivotally connected to the chest support plate **100**, thus defining an upper pivot point thereof.

(76) As depicted in FIGS. **11** and **12**, the rack and pinion mechanism **122** includes a drive screw **148** engaging the drive nut **124**. Coupled gears **150** are attached to the carriage **130**. The larger of the gears **150** engage an upper rack **152** (fixed within the offset main beam **12**), and the smaller of the gears **150** engage a lower rack **154**. The carriage **130** is defined as a gear assembly that floats between the two racks **152** and **154**.

(77) As depicted in FIGS. **11** and **12**, the rack and pinion mechanism **122** converts rotation of the drive screw **148** into linear translation of the first and second links **112** and **114** in the first and second elongated slots **120** and **128** toward the first portion **40** of the support structure **14**. As the drive nut **124** translates along drive screw **148** (via rotation of the drive screw **148**), the carriage **130** translates towards the first portion **40** with less travel due to the different gear sizes of the coupled gears **150**. The difference in travel, influenced by different gear ratios, causes the first links **112** pivotally attached thereto to lift the chest support plate **100**. Lowering of the chest support plate **100** is accomplished by performing this operation in reverse. The second links **114** are “idler” links (attached to the drive nut **124** and the chest support plate **100**) that controls the tilt of the chest support plate **100** as it is being lifted and lowered. All components associated with lifting while tilting the chest plate predetermine where COR **108** resides. Furthermore, a servomotor (not shown) interconnected with the drive screw **148** can be computer controlled and/or operated by the operator of the surgical frame **10** to facilitate controlled lifting and lowering of the chest support plate **100**. A safety feature can be provided, enabling the operator to read and limit a lifting and lowering force applied by the torso-lift support **24** in order to prevent injury to the patient P. Moreover, the torso-lift support **24** can also include safety stops (not shown) to prevent over-extension or compression of the patient P, and sensors (not shown) programmed to send patient position feedback to the safety stops.

(78) An alternative preferred embodiment of a torso-lift support is generally indicated by the

numeral **160** in FIGS. **13**A-**15**. As depicted in FIGS. **13**A-**13**C, an alternate offset main beam **162** is utilized with the torso-lift support **160**. Furthermore, the torso-lift support **160** has a support plate **164** pivotally linked to the offset main beam **162** by a chest support lift mechanism **166**. An arm support rod/plate **168** is connected to the support plate **164**, and the second arm support **22**B. The support plate **164** is attached to the chest support plate **100**, and the chest support lift mechanism **166** includes various actuators **170**A, **170**B, and **170**C used to facilitate positioning and repositioning of the support plate **164** (and hence, the chest support plate **100**).

(79) As discussed below, the torso-lift support **160** depicted in FIGS. **13**A-**15** enables a COR **172** thereof to be programmably altered such that the COR **172** can be a fixed COR or a variable COR. As their names suggest, the fixed COR stays in the same position as the torso-lift support **160** is actuated, and the variable COR moves between a first position and a second position as the torso-lift support **160** is actuated between its initial position and final position at full travel thereof. Appropriate placement of the COR **172** is important so that spinal cord integrity is not compromised (i.e., overly compressed or stretched). Thus, the support plate **164** (and hence, the chest support plate **100**) follows a path coinciding with a predetermined COR **172** (either fixed or variable). FIG. **13**A depicts the torso-lift support **160** retracted, FIG. **13**B depicts the torso-lift support **160** at half travel, and FIG. **13**C depicts the torso-lift support **160** at full travel.

(80) As discussed above, the chest support lift mechanism **166** includes the actuators **170**A, **170**B, and **170**C to position and reposition the support plate **164** (and hence, the chest support plate **100**). As depicted in FIGS. **14** and **15**, for example, the first actuator **170**A, the second actuator **170**B, and the third actuator **170**C are provided. Each of the actuators **170**A, **170**B, and **170**C are interconnected with the offset main beam **12** and the support plate **164**, and each of the actuators **170**A, **170**B, and **170**C are moveable between a retracted and extended position. As depicted in FIGS. **13**A-**13**C, the first actuator **170**A is pinned to the offset main beam **162** using a pin **174** and pinned to the support plate **164** using a pin **176**. Furthermore, the second and third actuators **170**B and **170**C are received within the offset main beam **162**. The second actuator **170**B is interconnected with the offset main beam **162** using a pin **178**, and the third actuator **170**C is interconnected with the offset main beam **162** using a pin **180**.

(81) The second actuator **170**B is interconnected with the support plate **164** via first links **182**, and the third actuator **170**C is interconnected with the support plate **164** via second links **184**. First ends **190** of the first links **182** are pinned to the second actuator **170**B and elongated slots **192** formed in the offset main beam **162** using a pin **194**, and first ends **200** of the second links **184** are pinned to the third actuator **170**C and elongated slots **202** formed in the offset main beam **162** using a pin **204**. The pins **194** and **204** are moveable within the elongated slots **192** and **202**. Furthermore, second ends **210** of the first links **182** are pinned to the support plate **164** using the pin **176**, and second ends **212** of the second links **184** are pinned to the support plate **164** using a pin **214**. To limit interference therebetween, as depicted in FIGS. **13**A-**13**C, the first links **182** are provided on the exterior of the offset main beam **162**, and, depending on the position thereof, the second links **184** are positioned on the interior of the offset main beam **162**.

(82) Actuation of the actuators **170**A, **170**B, and **170**C facilitates movement of the support plate **164**. Furthermore, the amount of actuation of the actuators **170**A, **170**B, and **170**C can be varied to affect different positions of the support plate **164**. As such, by varying the amount of actuation of the actuators **170**A, **1706**, and **170**C, the COR **172** thereof can be controlled. As discussed above, the COR **172** can be predetermined, and can be either fixed or varied. Furthermore, the actuation of the actuators **170**A, **170**B, and **170**C can be computer controlled and/or operated by the operator of the surgical frame **10**, such that the COR **172** can be programmed by the operator. As such, an algorithm can be used to determine the rates of extension of the actuators **170**A, **1706**, and **170**C to control the COR **172**, and the computer controls can handle implementation of the algorithm to provide the predetermined COR. A safety feature can be provided, enabling the operator to read and limit a lifting force applied by the actuators **170**A, **170**B, and **170**C in order to prevent injury to

the patient P. Moreover, the torso-lift support **160** can also include safety stops (not shown) to prevent over-extension or compression of the patient P, and sensors (not shown) programmed to send patient position feedback to the safety stops.

(83) FIGS. **16-23** depict portions of the sagittal adjustment assembly **28**. The sagittal adjustment assembly **28** can be used to distract or compress the patient's lumbar spine during or after lifting or lowering of the patient's torso by the torso-lift supports. The sagittal adjustment assembly **28** supports and manipulates the lower portion of the patient's body. In doing so, the sagittal adjustment assembly **28** is configured to make adjustments in the sagittal plane of the patient's body, including tilting the pelvis, controlling the position of the upper and lower legs, and lordosing the lumbar spine.

(84) As depicted in FIGS. **16** and **17**, for example, the sagittal adjustment assembly **28** includes the pelvic-tilt mechanism **30** for supporting the thighs and lower legs of the patient P. The pelvic-tilt mechanism **30** includes a thigh cradle **220** configured to support the patient's thighs, and a lower leg cradle **222** configured to support the patient's shins. Different sizes of thigh and lower leg cradles can be used to accommodate different sizes of patients, i.e., smaller thigh and lower leg cradles can be used with smaller patients, and larger thigh and lower leg cradles can be used with larger patients. Soft straps (not shown) can be used to secure the patient P to the thigh cradle **220** and the lower leg cradle **222**. The thigh cradle **220** and the lower leg cradle **222** are moveable and pivotal with respect to one another and to the offset main beam **12**. To facilitate rotation of the patient's hips, the thigh cradle **220** and the lower leg cradle **222** can be positioned anterior and inferior to the patient's hips.

(85) As depicted in FIGS. **18** and **25**, for example, a first support strut **224** and second support struts **226** are attached to the thigh cradle **220**. Furthermore, third support struts **228** are attached to the lower leg cradle **222**. The first support strut **224** is pivotally attached to the offset main beam **12** via a support plate **230** and a pin **232**, and the second support struts **226** are pivotally attached to the third support struts **228** via pins **234**. The pins **234** extend through angled end portions **236** and **238** of the second and third support struts **226** and **228**, respectively. Furthermore, the lengths of second and third support struts **226** and **228** are adjustable to facilitate expansion and contraction of the lengths thereof.

(86) To accommodate patients with different torso lengths, the position of the thigh cradle **220** can be adjustable by moving the support plate **230** along the offset main beam **12**. Furthermore, to accommodate patients with different thigh and lower leg lengths, the lengths of the second and third support struts **226** and **228** can be adjusted.

(87) To control the pivotal angle between the second and third support struts **226** and **228** (and hence, the pivotal angle between the thigh cradle **220** and lower leg cradle **222**), a link **240** is pivotally connected to a captured rack **242** via a pin **244**. The captured rack **242** includes an elongated slot **246**, through which is inserted a worm gear shaft **248** of a worm gear assembly **250**. The worm gear shaft **248** is attached to a gear **252** provided on the interior of the captured rack **242**. The gear **252** contacts teeth **254** provided inside the captured rack **242**, and rotation of the gear **252** (via contact with the teeth **254**) causes motion of the captured rack **242** upwardly and downwardly. The worm gear assembly **250**, as depicted in FIGS. **19-21**, for example, includes worm gears **256** which engage a drive shaft **258**, and which are connected to the worm gear shaft **248**.

(88) The worm gear assembly **250** also is configured to function as a brake, which prevents unintentional movement of the sagittal adjustment assembly **28**. Rotation of the drive shaft **258** causes rotation of the worm gears **256**, thereby causing reciprocal vertical motion of the captured rack **242**. The vertical reciprocal motion of the captured rack **242** causes corresponding motion of the link **240**, which in turn pivots the second and third support struts **226** and **228** to correspondingly pivot the thigh cradle **220** and lower leg cradle **222**. A servomotor (not shown) interconnected with the drive shaft **258** can be computer controlled and/or operated by the operator of the surgical frame **10** to facilitate controlled reciprocal motion of the captured rack **242**.

(89) The sagittal adjustment assembly **28** also includes the leg adjustment mechanism **32** facilitating articulation of the thigh cradle **220** and the lower leg cradle **222** with respect to one another. In doing so, the leg adjustment mechanism **32** accommodates the lengthening and shortening of the patient's legs during bending thereof. As depicted in FIG. **17**, for example, the leg adjustment mechanism **32** includes a first bracket **260** and a second bracket **262** attached to the lower leg cradle **222**. The first bracket **260** is attached to a first carriage portion **264**, and the second bracket **262** is attached to a second carriage portion **266** via pins **270** and **272**, respectively. The first carriage portion **264** is slidable within third portion **94** of the rear portion **74** of the offset main beam **12**, and the second carriage portion **266** is slidable within the first portion **90** of the rear portion **74** of the offset main beam **12**. An elongated slot **274** is provided in the first portion **90** to facilitate engagement of the second bracket **262** and the second carriage portion **266** via the pin **272**. As the thigh cradle **220** and the lower leg cradle **222** articulate with respect to one another (and the patient's legs bend accordingly), the first carriage **264** and the second carriage **266** can move accordingly to accommodate such movement.
(90) The pelvic-tilt mechanism **30** is movable between a flexed position and a fully extended position. As depicted in FIG. **22**, in the flexed position, the lumbar spine is hypo-lordosed. This opens the posterior boundaries of the lumbar vertebral bodies and allows for easier placement of any interbody devices. The lumbar spine stretches slightly in this position. As depicted in FIG. **23**, in the extended position, the lumbar spine is lordosed. This compresses the lumbar spine. When posterior fixation devices, such as rods and screws, are placed, optimal sagittal alignment can be achieved. During sagittal alignment, little to negligible angle change occurs between the thighs and the pelvis. The pelvic-tilt mechanism **30** also can hyper-extend the hips as a means of lordosing the spine, in addition to tilting the pelvis. One of ordinary skill will recognize, however, that straightening the patient's legs does not lordose the spine. Leg straightening is a consequence of rotating the pelvis while maintaining a fixed angle between the pelvis and the thighs.
(91) The sagittal adjustment assembly **28**, having the configuration described above, further includes an ability to compress and distract the spine dynamically while in the lordosed or flexed positions. The sagittal adjustment assembly **28** also includes safety stops (not shown) to prevent over-extension or compression of the patient, and sensors (not shown) programmed to send patient position feedback to the safety stops.
(92) As depicted in FIGS. **24-26**, for example, the coronal adjustment assembly **34** is configured to support and manipulate the patient's torso, and further to correct a spinal deformity, including but not limited to a scoliotic spine. As depicted in FIGS. **24-26**, for example, the coronal adjustment assembly **34** includes a lever **280** linked to an arcuate radiolucent paddle **282**. As depicted in FIGS. **24** and **25**, for example, a rotatable shaft **284** is linked to the lever **280** via a transmission **286**, and the rotatable shaft **284** projects from an end of the chest support plate **100**. Rotation of the rotatable shaft **284** is translated by the transmission **286** into rotation of the lever **280**, causing the paddle **282**, which is linked to the lever **280**, to swing in an arc. Furthermore, a servomotor (not shown) interconnected with the rotatable shaft **284** can be computer controlled and/or operated by the operator of the surgical frame **10** to facilitate controlled rotation of the lever **280**.
(93) As depicted in FIG. **24**, for example, adjustments can be made to the position of the paddle **282** to manipulate the torso and straighten the spine. As depicted in FIG. **25**, when the offset main beam **12** is positioned such that the patient P is positioned in a lateral position, the coronal adjustment assembly **34** supports the patient's torso. As further depicted in FIG. **26**, when the offset main beam **12** is positioned such that the patient P is positioned in a prone position, the coronal adjustment assembly **34** can move the torso laterally, to correct a deformity, including but not limited to a scoliotic spine. When the patient is strapped in via straps (not shown) at the chest and legs, the torso is relatively free to move and can be manipulated. Initially, the paddle **282** is moved by the lever **280** away from the offset main beam **12**. After the paddle **282** has been moved away from the offset main beam **12**, the torso can be pulled with a strap towards the offset main beam **12**.

The coronal adjustment assembly **34** also includes safety stops (not shown) to prevent over-extension or compression of the patient, and sensors (not shown) programmed to send patient position feedback to the safety stops.
(94) A preferred embodiment of a surgical frame incorporating a translating beam is generally indicated by the numeral **300** in FIGS. **27-30**. Like the surgical frame **10**, the surgical frame **300** serves as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby. In doing so, the surgical frame **300** serves to support the patient P such that the patient's spine does not experience unnecessary stress/torsion.
(95) The surgical frame **300** includes translating beam **302** that is generally indicated by the numeral **302** in FIGS. **27-30**. The translating beam **302** is capable of translating motion affording it to be positioned and repositioned with respect to portions of the remainder of the surgical frame **300**. As discussed below, the positioning and repositioning of the translating beam **302**, for example, affords greater access to a patient receiving area A defined by the surgical frame **300**, and affords greater access to the patient P by a surgeon and/or a surgical assistant (generally indicated by the letter S in FIG. **30**) via access to either of the lateral sides L.sub.1 and L.sub.2 (FIG. **30**) of the surgical frame **300**.
(96) As discussed below, by affording greater access to the patient receiving area A, the surgical frame **300** affords transfer of the patient P from and to a surgical table/gurney. Using the surgical frame **300**, the surgical table/gurney can be conventional, and there is no need to lift the surgical table/gurney over portions of the surgical frame **300** to afford transfer of the patient P thereto.
(97) The surgical frame **300** is configured to provide a relatively minimal amount of structure adjacent the patient's spine to facilitate access thereto and to improve the quality of imaging available before, during, and even after surgery. Thus, the workspace of a surgeon and/or a surgical assistant and imaging access are thereby increased. The workspace, as discussed below, can be further increased by positioning and repositioning the translating beam **302**. Furthermore, radiolucent or low magnetic susceptibility materials can be used in constructing the structural components adjacent the patient's spine in order to further enhance imaging quality.
(98) The surgical frame **300**, as depicted in FIGS. **27-30**, is similar to the surgical frame **10** except that surgical frame **300** includes a support structure **304** having a support platform **306** incorporating the translating beam **302**. The surgical frame **300** incorporates the offset main beam **12** and the features associated therewith from the surgical table **300**. As such, the element numbering used to describe the surgical frame **10** is also applicable to portions of the surgical frame **300**.
(99) Rather than including the cross member **44**, and the horizontal portions **46** and the vertical portions **48** of the first and second support portions **40** and **42**, the support structure **304** includes the support platform **306**, a first vertical support post **308**A, and a second vertical support post **308**B. As depicted in FIGS. **27-30**, the support platform **306** extends from adjacent one longitudinal end to adjacent the other longitudinal end of the surgical frame **300**, and the support platform **306** supports the first vertical support post **308**A at the one longitudinal end and supports the second vertical support post **308**B at the other longitudinal end.
(100) As depicted in FIGS. **27-30**, the support platform **306** (in addition to the translating beam **302**) includes a first end member **310**, a second end member **312**, a first support bracket **314**, and a second support bracket **316**. Casters **318** are attached to the first and second end members **310** and **312**. The first end member **310** and the second end member **312** each include an upper surface **320** and a lower surface **322**. The casters **318** can be attached to the lower surface of each of the first and second end members **310** and **312** at each end thereof, and the casters **318** can be spaced apart from one another to afford stable movement of the surgical frame **300**. Furthermore, the first support bracket **314** supports the first vertical support post **308**A, and the second support bracket **316** supports the vertical second support post **308**B.
(101) The translating beam **302** is interconnected with the first and second end members **310** and

**312** of the support platform **306**, and as depicted in FIGS. **27-30**, the translating beam **302** is capable of movement with respect to the first and second end members **310** and **312**. The translating beam **302** includes a first end member **330**, a second end member **332**, a first L-shaped member **334**, a second L-shaped member **336**, and a cross member **338**. The first L-shaped member **334** is attached to the first end member **330** and the cross member **338**, and the second L-shaped member **336** is attached to the second end member **332** and the cross member **338**. Portions of the first and second L-shaped members **334** and **336** extend downwardly relative to the first and second end members **330** and **332** such that the cross member **338** is positioned vertically below the first and second end member **330** and **332**. The vertical position of the cross member **338** relative to the remainder of the surgical frame **300** lowers the center of gravity of the surgical frame **300**, and in doing so, serves in adding to the stability of the surgical frame **300**.
(102) The translating beam **302**, as discussed above, is capable of being positioned and repositioned with respect to portions of the remainder of the surgical frame **300**. To that end, the support platform **306** includes a first translation mechanism **340** and a second translation mechanism **342**. The first translation mechanism **340** facilitates attachment between the first end members **310** and **330**, and the second translation mechanism **342** facilitates attachment between the second end members **312** and **332**. The first and second translation mechanism **340** and **342** also facilitate movement of the translating beam **302** relative to the first end member **310** and the second end member **312**.
(103) The first and second translation mechanisms **340** and **342** can each include a transmission **350** and a track **352** for facilitating movement of the translating beam **302**. The tracks **352** are provided on the upper surface **320** of the first and second end members **310** and **312**, and the transmissions **350** are interoperable with the tracks **352**. The first and second transmission mechanisms **340** and **342** can each include an electrical motor **354** or a hand crank (not shown) for driving the transmissions **350**. Furthermore, the transmissions **350** can include, for example, gears or wheels driven thereby for contacting the tracks **352**. The interoperability of the transmissions **350**, the tracks **352**, and the motors **354** or hand cranks form a drive train for moving the translating beam **302**. The movement afforded by the first and second translation mechanism **340** and **342** allows the translating beam **302** to be positioned and repositioned relative to the remainder of the surgical frame **300**.
(104) The surgical frame **300** can be configured such that operation of the first and second translation mechanism **340** and **342** can be controlled by an operator such as a surgeon and/or a surgical assistant. As such, movement of the translating beam **302** can be effectuated by controlled automation. Furthermore, the surgical frame **300** can be configured such that movement of the translating beam **302** automatically coincides with the rotation of the offset main beam **12**. By tying the position of the translating beam **302** to the rotational position of the offset main beam **12**, the center of gravity of the surgical frame **300** can be maintained in positions advantageous to the stability thereof.
(105) During use of the surgical frame **300**, access to the patient receiving area A and the patient P can be increased or decreased by moving the translating beam **302** between the lateral sides L.sub.1 and L.sub.2 of the surgical frame **300**. Affording greater access to the patient receiving area A facilitates transfer of the patient P between the surgical table/gurney and the surgical frame **300**. Furthermore, affording greater access to the patient P facilitates ease of access by a surgeon and/or a surgical assistant to the surgical site on the patient P.
(106) The translating beam **302** is moveable using the first and second translation mechanisms **340** and **342** between a first terminal position (FIG. **28**) and a second terminal position (FIGS. **29** and **30**). The translating beam **302** is positionable at various positions (FIG. **27**) between the first and second terminal positions. When the translating beam **302** is in the first terminal position, as depicted in FIG. **28**, the translating beam **302** and its cross member **338** are positioned on the lateral side L.sub.1 of the surgical frame **300**. Furthermore, when the translating beam **302** is in the

second terminal position, as depicted in FIGS. **29** and **30**, the translating beam **302** and its cross member **338** are positioned in the middle of the surgical frame **300**.
(107) With the translating beam **302** and its cross member **338** moved to be positioned at the lateral side L.sub.1, the surgical table/gurney and the patient P positioned thereon can be positioned under the offset main beam **12** in the patient receiving area A to facilitate transfer of the patient P to or from the offset main beam **12**. As such, the position of the translating beam **302** at the lateral side L.sub.1 enlarges the patient receiving area A so that the surgical table/gurney can be received therein to allow such transfer to or from the offset main beam **12**.
(108) Furthermore, with the translating beam **302** and its cross member **338** moved to be in the middle of the surgical frame **300** (FIGS. **29** and **30**), a surgeon and/or a surgical assistant can have access to the patient P from either of the lateral sides L.sub.1 or L.sub.2. As such, the position of the translating beam **302** in the middle of the surgical frame **300** allows a surgeon and/or a surgical assistant to get close to the patient P supported by the surgical frame **300**. As depicted in FIG. **30**, for example, a surgeon and/or a surgical assistant can get close to the patient P from the lateral side L.sub.2 without interference from the translating beam **302** and its cross member **338**. The position of the translating beam **302** can be selected to accommodate access by both a surgeon and/or a surgical assistant by avoiding contact thereof with the feet and legs of a surgeon and/or a surgical assistant.
(109) The position of the translating beam **302** and its cross member **338** can also be changed according to the rotational position of the offset main beam **12**. To illustrate, the offset main beam **12** can be rotated a full 360° before, during, and even after surgery to facilitate various positions of the patient to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the offset main beam **12** can be positioned by the surgical frame **300** to place the patient P in a prone position (e.g., FIGS. **27** and **28**), lateral positions (e.g., FIGS. **29** and **30**), and in a position 45° between the prone and lateral positions. The translating beam **302** can be positioned to accommodate the rotational position of the offset main beam **12** to aid in the stability of the surgical frame **300**. For example, when the patient P is in the prone position, the translating beam **302** can preferably be moved to the center of the surgical frame **300** underneath the patient P. Furthermore, when the patient P is in one of the lateral positions, the translating beam **302** can be moved toward one of the corresponding lateral sides L.sub.1 and L.sub.2 of the surgical frame **300** to position underneath the patient P. Such positioning of the translating beam **302** can serve to increase the stability of the surgical frame **300**.
(110) A portion of a surgical frame **400** incorporating an upper leg support **402** in accordance with a first embodiment of the present disclosure is described hereinbelow. The surgical frame **400** can incorporate the features of the above-discussed surgical frames, and the upper leg support **402** can also be incorporated in the above-discussed surgical frames. As discussed below, the upper leg support **402** is reconfigurable such reconfiguration can be done via articulation using manual adjustment or controlled automation of the componentry thereof. In addition to the upper legs of the patient P, the upper leg support **402** can be used to at least partially support the pelvic area of the patient P, and to facilitate manipulation of the lumbar spine of the patient P.
(111) Like the surgical frames **10** and **300**, the surgical frame **400** can serve as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby. In doing so, the surgical frame **400** serves to support the patient P such that the patient's spine does not experience unnecessary stress/torsion.
(112) Like the surgical frame **300**, the surgical frame **400** can include a translating beam **302** and a support structure **304** having a support platform **306** incorporating the translating beam **302**. Besides the support platform **306**, the support structure **304** can include a first vertical support portion **308**A and a second vertical support portion **308**B. The first vertical support portion **308**A and the second vertical support portion **308**B are capable of expansion and contraction.
(113) As depicted in FIGS. **31**-**33**B, the surgical frame **400** also incorporates a main beam **404**

having a first end (not shown) attached relative to the first support portion **308**A and a second end (not shown) attached relative to the second support portion **308**B. Like in the surgical frame **300**, the main beam includes a first portion (not shown) at the first end, a second portion (not shown) at the second end, and a third portion **410** extending between the first portion and the second portion. The main beam **404** is similar to the offset main beam **12**, and, as discussed below, the main beam **404** can incorporate features associated with the offset main beam **12**. To illustrate, the offset main beam **404**, like the main beam **12**, is used in supporting the patient P on the surgical frame **400** and includes various components similar to those incorporated in the surgical frames **10** and **300**. For example, in addition to the upper leg support **402**, the main beam **404** can incorporate a head support (not shown), a chest support (not shown), arm supports (not shown), and a lower leg support (not shown). The upper leg support **402**, the chest support, the arm supports, and the lower leg support can be attached to the third portion **410** of the main beam **404**.
(114) An operator such as a surgeon can control actuation of the various support components to manipulate the position of the patient's body. Soft straps (not shown) are used with these various support components to secure the patient P to the frame and to enable either manipulation or fixation of the patient P. Furthermore, reusable soft pads can be used on the load-bearing areas of the various support components. Additionally, the main beam **404** can be rotated a full 360° before, during, and even after surgery to facilitate various positions of the patient P to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the main beam **404** can be positioned by the surgical frame **400** to place the patient P in a prone position, lateral positions, and in a position 45° between the prone and lateral positions.
(115) The surgical frame **400** can be used to facilitate access to different parts of the spine of the patient P. In particular, the surgical frame **400** can be used to facilitate access to portions of the patient's lumbar spine. To illustrate, the patient P is simultaneously supported by the chest support and the upper leg support **402** on the main beam **404**, and uninterrupted access is provided to portions of the patient's lumbar spine by the positions of the chest support and the upper leg support **402**. The upper leg support **402** can be used to support the patient P during rotation of the main beam **404**, and articulation of the other componentry of the surgical frame **400**. Furthermore, the upper leg support **402**, as depicted in FIGS. **36** and **38**, is actuatable to facilitate positioning and repositioning thereof before, during, and after surgery to manipulate the patient P about an adjustable center of rotation CR located in and/or along a portion of the spine, including but not limited to the lumbar spine. As discussed below, the adjustable center of rotation CR is both adjustable to accommodate patients having different body sizes, and adjustable to facilitate, for example, flexing of the lumbar spine with the center of rotation CR location located above (posterior), within, or below (anterior) the lumbar spine to afford surgical access thereto with or without distribution or compression of the lumbar spine or portions thereof.
(116) The main beam **404** is moveably attached relative to the first vertical support post **308**A and the second vertical support post **308**B. Like those of the surgical frames **10** and **300**, the first vertical support post **308**A and the second vertical support post **308**B of the surgical frame **400** each include a clevis (not shown) supporting componentry facilitating rotation of the main beam **404**. In addition to the clevis, the first vertical support post **308**A includes a support block portion (not shown), a pin portion (not shown) pivotally attaching the support block portion to the clevis, and an axle portion (not shown) rotatably supported by the support block and interconnected to the first portion at the first end of the main beam **404**. The support block portion, via interaction of the pin portion with the clevis, is capable of pivotal movement relative to the clevis to accommodate different heights for the first vertical support post **308**A and the second vertical support post **308**B. And the main beam **404**, via interaction of the axle portion with the support block portion, is capable of rotational movement relative to the support block portion to accommodate rotation of the patient P supported by the main beam **404**.
(117) Furthermore, in addition to the clevis, the second vertical support post **308**B includes a

coupler (not shown) and a pin portion (not shown) pivotally attaching the coupler to the clevis. The coupler includes a base portion (not shown) that is pinned to the clevis with the pin portion, a body portion (not shown) that includes a transmission (not shown), a motor (not shown) that drives the transmission in the body portion, and a head portion (not shown) that is rotatable with respect to the body portion and driven rotationally by the transmission via the motor. The head portion is interconnected with the second portion at the second end of the main beam **404**, and the head portion (via the transmission and the motor) can rotate the main beam **404** a full 360° before, during, and even after surgery to facilitate various positions of the patient P.

(118) As depicted in FIGS. **31**-**34**B, the upper leg support **402** can be attached to and incorporated into portions of the third portion **410** of the main beam **404**. The upper leg support **402**, as depicted in FIGS. **32**B-**38**, can include a first arm portion **450**, a second arm portion **452**, and a platform portion **454**. The first arm portion **450** includes a first end portion **460** and a second end portion **462**, and the second arm portion **452** includes a first end portion **464** and a second end portion **466**. The first end portion **460** and the second end portion **462** of the first arm portion **450** are pivotally attached, respectively, to the main beam **404** and the first end portion **464** of the second arm portion **452**, and the first end portion **464** and the second end portion **466** of the second arm portion **452** are pivotally attached, respectively, to the second end portion **466** of the first arm portion **450** and the main beam **404**.

(119) The first arm portion **450** is extendable, and includes a base portion **470** that includes the first end portion **460** and an extendable portion **472** that includes the second end portion **462**. The extendable portion **472** is moveable inwardly and outwardly relative to the base portion **470**, and such inward and outward movement serves to pivot the first arm portion **450** relative to the main beam **404**, pivot the first arm portion **450** and the second arm portion **452** relative to one another, and pivot the second arm portion **452** relative to the main beam **404**. As discussed below, such pivotal movement serves in facilitating positioning and repositioning of the platform portion **454** relative to the main beam **404**. To illustrate, increasing the amount of extension of the extendable portion **472** relative to the base portion **470** moves the platform portion **454** away from the third portion **410** of the main beam **404**, and toward the first end and away from the second end.

(120) The third portion **410**, as depicted in FIGS. **32**B and **33**B, includes an interior cavity **480** defined by a first sidewall portion **482**, a second sidewall portion **484**, and a connecting-wall portion **486** joining the first sidewall portion **482** and the second sidewall portion **484** to one another. Portions of the upper leg support **402** are received within the interior cavity **480**. To illustrate, the first end portion **460** of the first arm portion **450** and the second end portion **466** of the second arm portion **452** are received with the cavity **480**.

(121) The first end portion **460** of the first arm portion **450**, as depicted in FIG. **32**B, includes an aperture **490** for receiving a pin **492** extending between the first sidewall portion **482** and the second sidewall portion **484** to facilitate pivotal attachment of the first arm portion **450** to the main beam **404**, and the second end portion **466** of the second arm portion **452** includes an aperture **494** for receiving a pin **496** extending between the first sidewall portion **482** and the second sidewall portion **484** to facilitate pivotal attachment of the second arm portion **452** to the main beam **404**. Furthermore, the second end portion **462** of the first arm portion **450** and the first end portion **464** of the second arm portion **452** can form a clevis-tang joint, wherein one of the second end portion **462** and the first end portion **464** is a clevis, and the other of the second end portion **462** and the first end portion **464** is a tang. As depicted in FIGS. **32**B and **33**B, the second end portion **462** is configured as a tang with an aperture **500** extending therethrough, and the first end portion **464** is configured as a clevis with apertures **502** extending therethrough. The aperture **500** and the apertures **502**, as depicted in FIGS. **32**B and **33**B, are configured to receive a pin **504** to facilitate pivotal attachment of the first arm portion **450** and the second arm portion **452** to one another.

(122) Additionally, the pin **504** is used in facilitating attachment of the platform portion **454** to the first arm portion **450** and the second arm portion **452**. As depicted in FIGS. **32**B and **33**B, the

platform portion **454** includes a base portion **510**, a first upstanding portion **512**, a second upstanding portion **514**, and a third upstanding portion **516**. Portions of the first upstanding portion **512** and the second upstanding portion **514** can be received in the interior cavity **480**, and portions of the second end portion **462** of the first arm portion **450** and the first end portion **464** of the second arm portion **452** are received between the first upstanding portion **512** and the second upstanding portion **514**. The first upstanding portion **512** includes an aperture **520**, the second upstanding portion **514** includes an aperture **522**, and each of the first aperture **520** and the second aperture **522** are configured to receive portions of the pin **504** to attach the second end portion **462** of the first arm portion **450** and the first end portion **464** of the second arm portion **452** to platform portion **454**.

(123) The extension of the extendable portion **472** relative to the base portion **470** serves in pivoting the first arm portion **450** and the second arm portion **452** relative to one another such that increasing the amount of extension decreases an angle between the first arm portion **450** and the second arm portion **452**, and decreasing the amount of extension increases the angle between the first arm portion **450** and the second arm portion **452**. Given that the platform portion **454** is attached to the second end portion **462** of the first arm portion **450** and the first end portion **464** of the second arm portion **452**, increasing the amount of extension of the first arm portion **450** moves the platform portion **454** away from the third portion **410** of the main beam **404**, and toward the first end and away from the second end, and decreasing the amount of extension of the first arm portion **450** moves the platform portion **454** toward the third portion **410** of the main beam **404**, and away from the first end and toward the second end. Furthermore, when the extension of the first arm portion **450** is decreased, portions of the first upstanding portion **512** and the second upstanding portion **514** are drawn into the cavity **480**. As discussed below, the movement of the platform portion **454** using the extension of the extendable portion **472** ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame **400**.

(124) As depicted in FIGS. **32**B and **33**B, the upper leg support **402** also includes a telescoping shaft portion **530** that is connected between the second arm portion **452** and the platform portion **454**. The telescoping shaft portion **530** is used to pivot the platform portion **454** relative to the first arm portion **450** and the second arm portion **452**. The telescoping shaft portion **530** includes a first end portion **532**, a second end portion **534**, a base portion **536** including the first end portion **532**, and an extendable portion **538** including the second end portion **534**. As discussed below, the extendable portion **538** is moveable inwardly and outwardly relative to the base portion **536**, and such inward and outward movement serves to pivot the platform portion **454**.

(125) To facilitate connection between the telescoping shaft portion **530** and the second arm portion **452**, one of the first end portion **532** and the second arm portion **452** can form a clevis, and the other of the first end portion **532** and the second arm portion **452** can form a tang. Furthermore, to facilitate connection between the telescoping shaft portion **530** and the platform portion **454**, one of the second end portion **534** and the platform portion **454** can form a clevis, and the other of the second end portion **534** and the platform portion **454** can form a tang. As depicted in FIGS. **32**B and **33**B, the second arm portion **452** includes a clevis **540** having apertures **542**, the first end portion **532** is used as a tang having an aperture **544**, and a pin **546** is received through the apertures **542** and **544** to join the telescoping shaft portion **530** to the second arm portion **542**. Furthermore, as depicted in FIGS. **32**B and **33**B, the platform portion **454** includes a post **550**, the second end portion **534** includes an aperture **552**, and the post **550** is received in the aperture **552** to join the telescoping shaft portion **530** to the platform portion **454**.

(126) The extendable portion **538** is moveable inwardly and outwardly relative to the base portion **536**, and such inward and outward movement serves to pivot the platform portion **454** relative to the first arm portion **450** and the second arm portion **452**. Such pivotal movement serves in facilitating positioning and repositioning of the platform portion **454** relative to first arm portion

**450** and the second arm portion **452**. To illustrate, the base portion **510** of the platform portion **554** includes a first end **560** and a second end **562**, and increasing the amount of extension of the extendable portion **538** relative to the base portion **536** moves the second end **562** away from the third portion **410** of the main beam **404**. As discussed below, the movement of the platform portion **454** using the extension of the telescoping shaft portion **530** ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame **400**.

(127) As depicted in FIGS. **34**A and **34**B, the upper leg support **402** also includes a linear movement assembly **570**. The linear movement assembly **570** includes a track portion **572** attached to the third upstanding portion **516**, two trucks **574** moveable along the track portion **572**, and a support bracket **576** attached to the two trucks **574**. The third upstanding portion **516**, as depicted in FIGS. **33**A and **33**B, includes a first end **580** and a second end **582**, and is larger than the first upstanding portion **512** and the second upstanding portion **514**. The third upstanding portion **516** is attached at and adjacent the first end **580** to the base portion **510**, and extends from the base portion **510** toward the first portion of the main beam **404** to the second end **582**. The third upstanding portion **516** supports the track portion **572**, the two trucks **574**, the support bracket **576**, and additional components of the upper leg support **402**.

(128) The linear movement assembly **570**, as depicted in FIGS. **34**A and **34**B, also includes a telescoping shaft portion **590** that is connected between the third upstanding portion **516** and the support bracket **576**. The telescoping shaft portion **590** includes a first end portion **592**, a second end portion **594**, a base portion **596** including the first end portion **592**, and an extendable portion **598** including the second end portion **594**. To attach the telescoping shaft portion **590** to the third upstanding portion **516**, the first end portion **592** of the telescoping shaft portion **590** can include an aperture **600**, the third upstanding portion **516** can include an aperture **602**, and a pin **604** can be received in the apertures **600** and **602**. Furthermore, to attach the telescoping shaft portion **590** to the support bracket **576**, the second end portion **594** of the telescoping shaft portion **590** can include an aperture **606**, the support bracket **576** can include a projection **608** including an aperture **610**, and a pin **612** can be received in the apertures **606** and **610**.

(129) The extendable portion **598** is moveable inwardly and outwardly relative to the base portion **596**, and such inward and outward movement relative to the base portion **596** serves to move the support bracket **576** via movement of the two trucks **574** along the track portion **572** between at least a first position closer to the second end **582** of the main beam **404** to a second position closer to the first end **580** of the main beam **404**. As discussed below, the movement of the support bracket **576** using the extension of the extendable portion **598** ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame **400**.

(130) The upper leg support **402** also includes a support assembly **620** that is carried by the support bracket **576**. The support assembly **620** includes a first support post **622**, a second support post **624**, and a connecting bracket **626** connecting the first support post **622** and the second support post **624** to one another. The support assembly **620** also includes a first support block **630**, a second support block **632**, a third support block **634**, a fourth support block **636**, a first support plate **640**, a second support plate **642**, and a third support plate **644**. Each of the first support plate **640**, the second support plate **642**, and the third support plate **644** include upper surfaces **646**A, **646**B, and **646**C, respectively, that can be used to contact the upper legs of the patient. The upper surfaces **646**A and **646**C can be covered with padding (not shown) for contacting portions of the patient's upper legs, and the padding can include pressure sensors (not shown) incorporated therein. The resulting pressure sensing padding can be used to determine if undue stress is placed on the patient P via articulation of the upper leg support **402**.

(131) The first support plate **640** and the second support plate **642**, as discussed below, are moveable with respect to the support bracket **576** and the third support plate **644**. As depicted in

FIGS. **33**A and **33**B, the third support plate **644** is attached to the support bracket **576**, and the second support plate **642** is positioned such that it can move underneath the third support plate **644**. As such, during movement of the first support plate **640** and the second support plate **642**, the area defined by the upper surfaces **646**A, **646**B, and **646**C can be respectively decreased or increased as the second support plate **642** is moved under or out from under the third support plate **644**.
(132) As depicted in FIG. **34**A, the first support plate **640** is attached to the first support block **630** and the second support block **632**, and the first support block **630** is moveable along the first support post **622** and the second support block **632** is moveable along the second support post **624**. Furthermore, the second support plate **642** is attached to the third support block **634** and the fourth support block **636**, and the third support block **634** is moveable along the first support post **622** and the fourth support block **636** is moveable along the second support post **624**.
(133) The first and second support blocks **630** and **632** include apertures **650** and **652** for receiving the first and second support posts **622** and **624**, respectively, and the third and fourth support blocks **634** and **636** include apertures **654** and **656** for receiving the first second support posts **622** and **624**, respectively. The first support plate **640** and the second support plate **642** are moveable inwardly and outwardly relative to the support bracket **576** and the third support plate **644** with threads complementary to those of the threaded shaft via movement of the first and third support blocks **630** and **634** on the first support post **622** and via movement of the second and fourth support blocks **632** and **636** on the second support post **624**.
(134) The support assembly **620** also includes a threaded shaft **660** that is retained in position between the support bracket **576** and the connecting bracket **626**. As discussed below, the threaded shaft **660** is used to constrain movement of the first support plate **640** and the second support plate **642** relative to the third support plate **644** and the main beam **404**.
(135) Furthermore, the first support plate **640** includes a first support collar **664** opposite from the upper surface **646**A, and the second support plate **642** includes a second support collar **666** opposite from the upper surface **646**B. The first support collar **664** includes a first aperture **670** that can include threads complementary to those of the threaded shaft **660**, and the second support collar **666** includes a second aperture **672** that can include threads complementary to those of the threaded shaft **660**. The threaded shaft **660** can be received in the first aperture **670** and the second aperture **672**. The first support collar **664** and the second support collar **666** can include one or more latches (not shown) that can be engaged and disengaged from the threaded shaft **660** via actuation thereof. The one or more latches can be attached to the first support collar **664** and/or the second support collar **666**, and a user can actuate the one or more latches to engage or disengage the threaded shaft **660** to correspondingly prevent movement or allow movement of the first support collar **664** and the second support collar **666** along the threaded shaft **660**. When the one or more latches are engaged, the interactions of the one or more latches with the threaded shaft **660** prevent movement of the first support plate **640** and the second support plate **642** relative to the third support plate **644**. When the one or more latches are disengaged, the first support plate **640** and the second support plate **642** can move relative to the third support plate **644**. Rather than using the threaded shaft **660**, a shaft with catches and/or teeth to which the one or more latches can be engaged and disengaged.
(136) Alternatively, a motor/transmission/actuator (not shown) can be used to facilitate rotation of the threaded shaft **660**, and rotation of the threaded shaft **660** and the interaction in the first aperture **670** and the second aperture **672** causes corresponding movement of the first support plate **640** and the second support plate **642**. As such, rotation of the threaded shaft **660** via actuation of the motor/transmission/actuator is translated into movement of the first support plate **640** and the second support plate **642**. To illustrate, the threads of the threaded shaft **660**, the first aperture **670**, and the second aperture **672** can be configured such that clockwise rotation of the threaded shaft **660** via actuation of the motor/transmission/actuator causes inward movement of the first support plate **640** and the second support plate **642**, and counterclockwise rotation of the threaded shaft **660**

via actuation of the motor/transmission/actuator causes outward movement of the first support plate **640** and the second support plate **642**. The inward and outward movement of the first support plate **640** and the second support plate **642** is relative to the third support plate **644** and the main beam **404**.

(137) The movement of the first support plate **640** and the second support plate **642** ultimately serves in adjusting a total width of a combination of the first support plate **640**, the second support plate **642**, and the third support plate **644**. Adjustment of the combined width of the first support plate **640**, the second support plate **642**, and the third support plate **644** affords the accommodation of differently sized patients on the upper leg support **402**.

(138) The movement of the componentry of the upper leg support **402** can be effectuated via manual adjustment and/or controlled automation. To illustrate, the first arm portion **450** includes the extendable portion **472** that is moveable with respect to the base portion **470** thereof, the telescoping shaft portion **530** includes the extendable portion **538** that is moveable with respect to the base portion **536** thereof, the telescoping shaft portion **590** includes the extendable portion **598** that is moveable with respect to the base portion **596** thereof, and the motor/transmission/actuator facilitates movement of the first support plate **640** and the second support plate **642** is relative to the third support plate **644** and the main beam **404**.

(139) Such reconfiguration of the upper leg support **402** can be actuated using the manual adjustment and/or the controlled automation, and as discussed below, the extension and retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598**, as depicted in FIGS. **35** and **37**, for example, via such actuation can be used to both adjust the adjustable center of rotation CR to accommodate patients having different body sizes, and to facilitate flexing of the lumbar spine to afford surgical access thereto via the manipulation of portions thereof. The extension/retraction of the extendable portion **472** serves to change the angle of the first arm portion **450** and the second arm portion **452** relative to one another, the extension/retraction of the extendable portion **538** serves to change the angle of the platform portion **454** relative to the second arm portion **452**, and the extendable portion **598** serves to change position of the bracket **576** (which supports the first support plate **640**, the second support plate **642**, and the third support plate **644**) relative to the platform portion **454**.

(140) Using the upper leg support **402**, the position of the patient's upper legs can be altered, which correspondingly affects the flexure of the lumbar spine of the patient P, and care should be taken to prevent unwanted torsion thereof when manipulating the patient's spine. To illustrate, the amounts of extension/retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598** can be constrained with respect to one another to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support **402**. As such, the amounts of extension/retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598** can be contingent upon one another to facilitate such approximate preservation.

(141) A controller (not shown) with a user interface (not shown) can be used to control the constrained/contingent extension and/or retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598** via the controlled automation. Furthermore, because patients' heights can vary, the amounts of extension/retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598** can be altered to accommodate these different heights while still being constrained/contingent upon one another to provide for the desired amount of distraction/compression of portions of the lumbar spine during articulation of the upper leg support **402**.

(142) The controller with input via the user interface can allow the user to select the desired center of rotation and the desired amount of manipulation or angulation of the segmental portions of the lumbar spine. To illustrate, the user interface can be used to display a graphical or actual representation of the patient's spine, and the user interface can permit the user to input the desired center of rotation and the desired amount of manipulation by, for example, highlighting a portion of

the graphical or actual representation of the patient's spine on the user interface. The selection of the desired amount of manipulation can allow the user to select where the forces applied via the actuation of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598** are applied during flexure of the patient's spine. In addition to or alternatively to use of the user interface, a navigation tool interconnected with the controller and/or the user interface can be positioned on or adjacent the patient's spine to facilitate inputting of the desired center of rotation and the desired amount of manipulation. The inputting of the desired center of rotation and the desired amount of manipulation can be done with the main beam **404** and the patient P supported on the main beam **404** in various rotational positions including, but not limited to, prone, lateral, and supine positions.

(143) When the upper leg support **402** is articulated such that the lumbar spine of the patient P is in an unflexed neutral position, as depicted in FIG. **36**, the controller can be used to extend/retract the extendable portion **472** (of the first arm portion **450**), the extendable portion **538** (of the telescoping shaft portion **530**), and the extendable portion **598** (of the telescoping shaft portion **590**) such that the first arm portion **450**, the telescoping shaft portion **530**, and the telescoping shaft portion **590** have lengths of 24.862, 8.639, and 11.963 inches, respectively, that accommodate the height of the patient P. Furthermore, when the upper leg support **402** is articulated such that the lumbar spine of the patient P has a 30 degree flex, as depicted in FIG. **38**, the controller can be used to extend/retract the extendable portion **472** (of the first arm portion **450**), the extendable portion **538** (of the telescoping shaft portion **530**), and the extendable portion **598** (of the telescoping shaft portion **590**) such that the first arm portion **450**, the telescoping shaft portion **530**, and the telescoping shaft portion **590** have lengths of 40.375, 6.652, and 9.540 inches, respectively, that flex the lumbar spine of the patient P to afford surgical access thereto. Moreover, during the transition between the positions of FIGS. **36** and **38**, the controller can serve to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support **402** by properly adjusting the amounts of extension/retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598**.

(144) FIG. **39**, for example, includes a table that illustrates the relative amounts of increase/decrease of the lengths of the first arm portion **450** (Cylinder 1), the telescoping shaft portion **530** (Cylinder 2), and the telescoping shaft portion **590** (Cylinder 3) via extension/retraction of the extendable portion **472**, the extendable portion **538**, and the extendable portion **598** for a patient P having a height of 6′3″. As such, the upper leg support **402** provides an adjustable center of rotation CR located in the lumbar spine to accommodate patients having different body sizes, and also to afford surgical access to the lumbar spine via manipulation of portions thereof.

(145) Thus, using the user interface of the controller, the operator of the surgical frame **400** can input the height of the patient P, and input the desired degree of flexure of the lumbar spine, and the controller can actuate the first arm portion **450** (to extend or retract the extendable portion **472**), the telescoping shaft portion **530** (to extend or retract the extendable portion **538**), and the telescoping shaft portion **590** (to extend or retract the extendable portion **598**) the appropriate amounts to provide such flexion, while also preventing unwanted torsion of the patient's spine. As discussed above, the extension/retraction of the extendable portion **472** serves to change the angle of the first arm portion **450** and the second arm portion **452** relative to one another, the extension/retraction of the extendable portion **538** serves to change the angle of the platform portion **454** relative to the second arm portion **452**, and the extendable portion **598** serves to change position of the bracket **576** (which supports the first support plate **640**, the second support plate **642**, and the third support plate **644**) relative to the platform portion **454**. During such manipulation of the patient's spine using the upper leg support **402**, the lengths of the first arm portion **450**, the telescoping shaft portion **530**, and the telescoping shaft portion **590** may each alternatingly increase/decrease or decrease/increase to provide for the desired adjustable center of rotation CR. The operator can use

the controller to manipulate the upper leg support **402** to flex the lumbar spine of the patient P into position for surgery, while simultaneously inhibiting the unwanted torsion of the patient's spine that may be caused by reconfiguration of the upper leg support **402**.

(146) A portion of a surgical frame **1400** incorporating an upper leg support **1402** in accordance with a second embodiment of the present disclosure is described hereinbelow. The surgical frame **1400** can incorporate the features of the above-discussed surgical frames, and the upper leg support **1402** can also be incorporated in the above-discussed surgical frames. As discussed below, the upper leg support **1402** is reconfigurable such reconfiguration can be done via articulation using manual adjustment or controlled automation of the componentry thereof. In addition to the upper legs of the patient P, the upper leg support **1402** can be used to at least partially support the pelvic area of the patient P, and to facilitate manipulation of the lumbar spine of the patient P.

(147) Like the surgical frames **10** and **300**, the surgical frame **1400** can serve as an exoskeleton to support the body of the patient P as the patient's body is manipulated thereby. In doing so, the surgical frame **1400** serves to support the patient P such that the patient's spine does not experience unnecessary stress/torsion.

(148) Like the surgical frame **300**, the surgical frame **1400** can include a translating beam **302** and a support structure **304** having a support platform **306** incorporating the translating beam **302**. Besides the support platform **306**, the support structure **304** can include a first vertical support portion **308**A and a second vertical support portion **308**B. The first vertical support portion **308**A and the second vertical support portion **308**B are capable of expansion and contraction.

(149) As depicted in FIGS. **40-44**, the surgical frame **1400** also incorporates a main beam **1404** having a first end **1405**A attached relative to the first support portion **308**A and a second end **1405**B attached relative to the second support portion **308**A. Like in the surgical frame **300**, the main beam includes a first portion **1406** at the first end **1405**A, a second portion (not shown) at the second end **1405**B, and a third portion **1410** extending between the first portion **1406** and the second portion. The main beam **1404** is similar to the offset main beam **12**, and, as discussed below, the main beam **1404** can incorporate features associated with the offset main beam **12**. To illustrate, the offset main beam **1404**, like the main beam **12**, is used in supporting the patient P on the surgical frame **1400** and includes various components similar to those incorporated in the surgical frames **10** and **300**. For example, in addition to the upper leg support **1402**, the main beam **1404** can incorporate a head support (not shown), a chest support **1412**, arm supports (not shown), and a lower leg support **1416**. The upper leg support **1402**, the chest support **1412**, the arm supports, and the lower leg support **1416** can be attached to the third portion **1410** of the main beam **1404**.

(150) An operator such as a surgeon can control actuation of the various support components to manipulate the position of the patient's body. Soft straps (not shown) are used with these various support components to secure the patient P to the frame and to enable either manipulation or fixation of the patient P. Furthermore, reusable soft pads can be used on the load-bearing areas of the various support components. Additionally, the main beam **1404** can be rotated a full 360° before, during, and even after surgery to facilitate various positions of the patient P to afford various surgical pathways to the patient's spine depending on the surgery to be performed. For example, the main beam **1404** can be positioned by the surgical frame **1400** to place the patient P in a prone position, lateral positions, and in a position 45° between the prone and lateral positions.

(151) The surgical frame **1400** can be used to facilitate access to different parts of the spine of the patient P. In particular, the surgical frame **1400** can be used to facilitate access to portions of the patient's lumbar spine. To illustrate, the patient P is simultaneously supported by the chest support **1412** and the upper leg support **1402** on the main beam **1404**, and uninterrupted access is provided to portions of the patient's lumbar spine by the positions of the chest support **1412** and the upper leg support **1404**. The upper leg support **402** can be used to support the patient P during rotation of the main beam **1404**, and articulation of the other componentry of the surgical frame **1400**. Furthermore, the upper leg support **1402** is actuatable to facilitate positioning and repositioning of

the patient P before, during, and after surgery to manipulate the patient P about an adjustable center of rotation CR located in the lumbar spine. As discussed below, the adjustable center of rotation CR is both adjustable to accommodate patients having different body sizes, and adjustable to facilitate flexing of the lumbar spine to facilitate surgical access thereto via the manipulation of portions thereof.

(152) The main beam **1404** is moveably attached relative to the first vertical support post **308**A and the second vertical support post **308**B. Like those of the surgical frames **10** and **300**, the first vertical support post **308**A and the second vertical support post **308**B of the surgical frame **1400** each include a clevis (not shown) supporting componentry facilitating rotation of the main beam **1404**. In addition to the clevis, the first vertical support post **308**A includes a support block portion (not shown), a pin portion (not shown) pivotally attaching the support block portion to the clevis, and an axle portion (not shown) rotatably supported by the support block and interconnected to the first portion **1406** at the first end **1405**A of the main beam **1404**. The support block portion, via interaction of the pin portion with the clevis, is capable of pivotal movement relative to the clevis to accommodate different heights for the first vertical support post and the second vertical support post. And the main beam **1404**, via interaction of the axle portion with the support block portion, is capable of rotational movement relative to the support block portion to accommodate rotation of the patient P supported by the main beam **1404**.

(153) Furthermore, in addition to the clevis, the second vertical support post **308**B includes a coupler (not shown) and a pin portion (not shown) pivotally attaching the coupler to the clevis. The coupler includes a base portion (not shown) that is pinned to the clevis with the pin portion, a body portion (not shown) that includes a transmission (not shown), a motor (not shown) that drives the transmission in the body portion, and a head portion (not shown) that is rotatable with respect to the body portion and driven rotationally by the transmission via the motor. The head portion is interconnected with the second portion at the second end **1405**B of the main beam **1404**, and the head portion (via the transmission and the motor) can rotate the main beam **1404** a full 360° before, during, and even after surgery to facilitate various positions of the patient P.

(154) As depicted in FIGS. **40-46**, the upper leg support **1402** can be attached to and incorporated into portions of the third portion **1410** of the main beam **1404**. The upper leg support **1402**, as depicted in FIGS. **42-46**, can include a first arm portion **1450**, a second arm portion **1452**, and a platform portion **1454**. The first arm portion **1450** includes a first end portion **1460** and a second end portion **1462**, and the second arm portion **1452** includes a first end portion **1464** and a second end portion **1466**. The first end portion **1460** and the second end portion **1462** of the first arm portion **1450** are pivotally attached, respectively, to the main beam **1404** and the first end portion **1464** of the second arm portion **1452**, and the first end portion **1464** and the second end portion **1466** of the second arm portion **1452** are pivotally attached, respectively, to the second end portion **1466** of the first arm portion **1450** and the main beam **1404**.

(155) The first arm portion **1450** is extendable, and includes a base portion **11470** that includes the first end portion **1460** and an extendable portion **1472** that includes the second end portion **1462**. The extendable portion **1472** is moveable inwardly and outwardly relative to the base portion **1470**, and such inward and outward movement serves to pivot the first arm portion **1450** relative to the main beam **1404**, pivot the first arm portion **1450** and the second arm portion **1452** relative to one another, and pivot the second arm portion **1452** relative to the main beam **1404**. As discussed below, such pivotal movement serves in facilitating positioning and repositioning of the platform portion **1454** relative to the main beam **1404**. To illustrate, increasing the amount of extension of the extendable portion **1472** relative to the base portion **1470** moves the platform portion **1454** away from the third portion **1410** of the main beam **1404**, and toward the first end **1405**A and away from the second end **1405**B.

(156) The third portion **1410** includes an interior cavity **1890** defined by a first sidewall portion **1482**, a second sidewall portion **1484**, and a connecting-wall portion **1486** joining the first sidewall

portion **1482** and the second sidewall portion **1484** to one another. As depicted in FIGS. **42**, **44**, and **45**, portions of the upper leg support **1402** are received within the interior cavity **1480**. To illustrate, the first end portion **1460** of the first arm portion **1450** and the second end portion **1466** of the second arm portion **1452** are received with the cavity **1480**.

(157) The first end portion **1460** of the first arm portion **1450** includes an aperture **1490** for receiving a pin **1492** extending between the first sidewall portion **1482** and the second sidewall portion **1484** to facilitate pivotal attachment of the first arm portion **1450** to the main beam **1404**, and the second end portion **1466** of the second arm portion **1452** includes an aperture **1494** for receiving a pin **1496** extending between the first sidewall portion **1482** and the second sidewall portion **1484** to facilitate pivotal attachment of the second arm portion **1452** to the main beam **1404**. Furthermore, the second end portion **1462** of the first arm portion **1450** and the first end portion **1464** of the second arm portion **1452** can form a clevis-tang joint, wherein one of the second end portion **1462** and the first end portion **1464** is a clevis, and the other of the second end portion **1462** and the first end portion **1464** is a tang. As depicted in FIG. **43**, the second end portion **1462** is configured as a tang with an aperture **1500** extending therethrough, and the first end portion **1464** is configured as a clevis with apertures **1502** extending therethrough. The aperture **1500** and the apertures **1502** are configured to receive a pin **1504**, as depicted in FIG. **42**, to facilitate pivotal attachment of the first arm portion **1450** and the second arm portion **1452** to one another.

(158) Additionally, a pin **1506** is used in facilitating attachment of the platform portion **1454** to the second arm portion **1452**. As depicted in FIGS. **42** and **45**, the platform portion **1454** includes a base portion **1510**, a first upstanding portion **1512**, a second upstanding portion **1514**, and a third upstanding portion **1516**. Portions of the second end portion **1462** of the first arm portion **1450** and the first end portion **1464** of the second arm portion **1450** are received between the first upstanding portion **1512** and the second upstanding portion **1514**. The first upstanding portion **1512** includes an aperture **1520**, the second upstanding portion **1514** includes an aperture **1522**, and each of the first aperture **1520** and the second aperture **1522** are configured to receive portions of the pin **1506** to attach the first end portion **1464** of the second arm portion **1452** to platform portion **1454**.

(159) The extension of the extendable portion **1472** relative to the base portion **1470** serves in pivoting the first arm portion **1450** and the second arm portion **1452** relative to one another such that increasing the amount of extension decreases an angle between the first arm portion **1450** and the second arm portion **1452**, and decreasing the amount of extension increases the angle between the first arm portion **1450** and the second arm portion **1452**. Given that the platform portion **1454** is attached to the first end portion **1464** of the second arm portion **1452**, increasing the amount of extension of the first arm portion **1450** moves the platform portion **1454** away from the third portion **1410** of the main beam **1404**, and toward the first end **1405**A and away from the second end **1405**B, and decreasing the amount of extension of the first arm portion **1450** moves the platform portion **1454** toward the third portion **1410** of the main beam **1404**, and away from the first end **1405**A and toward the second end **14056**. As discussed below, the movement of the platform portion **1454** using the extension of the extendable portion **1472** ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame **1400**.

(160) As depicted in FIGS. **42-45**, the upper leg support **1402** also includes a telescoping shaft portion **1530** that is connected between the second arm portion **1452** and the platform portion **1454**. The telescoping shaft portion **1530** is used to pivot the platform portion **1454** relative to the second arm portion **1452**. The telescoping shaft portion **1530** includes a first end portion **1532**, a second end portion **1534**, a base portion **1536** including the first end portion **1532**, and an extendable portion **1538** including the second end portion **1534**. As discussed below, the extendable portion **1538** is moveable inwardly and outwardly relative to the base portion **1536**, and such inward and outward movement serves to pivot the platform portion **1454**.

(161) To facilitate connection between the telescoping shaft portion **1530** and the second arm portion **1452**, one of the first end portion **1532** and the second arm portion **1452** can form a clevis, and the other of the first end portion **1532** and the second arm portion **1452** can form a tang. Furthermore, to facilitate connection between the telescoping shaft portion **1530** and the platform portion **1454**, one of the second end portion **1534** and the platform portion **1454** can form a clevis, and the other of the second end portion **1534** and the platform portion **1454** can form a tang. As depicted in FIGS. **42** and **45**, the second arm portion **1452** includes a clevis **1540** having apertures **1542**, the first end portion **1532** is used as a tang having an aperture **1544**, and a pin **1546** is received through the apertures **1542** and **1544** to join the telescoping shaft portion **1530** to the second arm portion **1542**. Furthermore, as depicted in FIGS. **42** and **45**, the platform portion **1454** includes a clevis **1550** having apertures **1552**, the second end portion **1534** is used as a tang having an aperture **1554**, and a pin **1556** is received through the apertures **1552** and **1554** to join the telescoping shaft portion **1530** to the platform portion **1454**.
(162) The extendable portion **1538** is moveable inwardly and outwardly relative to the base portion **1536**, and such inward and outward movement serves to pivot the platform portion **1454** relative to the second arm portion **1452**. Such pivotal movement serves in facilitating positioning and repositioning of the platform portion **1454** relative to the second arm portion **1452**. To illustrate, the base portion **1510** of the platform portion **1454** includes a first end **1560** and a second end **1562**, and increasing the amount of extension of the extendable portion **1538** relative to the base portion **1536** moves the second end **1562** away from the third portion **1410** of the main beam **1404**. As discussed below, the movement of the platform portion **1454** using the extension of the telescoping shaft portion **1530** ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame **1400**.
(163) As depicted in FIGS. **45** and **46**, the upper leg support **1402** also includes a linear movement assembly **1570**. The linear movement assembly **1570** includes a track portion **1572** attached to the third upstanding portion **1516**, two trucks **1574** moveable along the track portion **1572**, and a support bracket **1576** attached to the two trucks **1574**. The third upstanding portion **1516**, as depicted in FIG. **46**, includes a first end **1580** and a second end **1582**, and is larger than the first upstanding portion **1512** and the second upstanding portion **1514**. The third upstanding portion **1516** is attached at and adjacent the first end **1580** to the base portion **1510**, and extends from the base portion **510** toward the first portion **1406** of the main beam **1404** to the second end **1582**. The third upstanding portion **1516** supports the track portion **1572**, the two trucks **1574**, the support bracket **1576**, and additional components of the upper leg support **1402**.
(164) The linear movement assembly **1570** also includes a telescoping shaft portion **1590** that is connected between the third upstanding portion **1516** and the support bracket **1576**. The telescoping shaft portion **1590** includes a first end portion **1592**, a second end portion **1594**, a base portion **1596** including the first end portion **1592**, and an extendable portion **1598** including the second end portion **1594**. To attach the telescoping shaft portion **1590** to the third upstanding portion **1516**, the first end portion **1592** of the base portion **1596** can include an aperture **1600**, the third upstanding portion **1516** can include a clevis **1602** with apertures **1603**, and a pin **1604** can be received in the apertures **1600** and **1603**. Furthermore, to attach the telescoping shaft portion **1590** to the support bracket **1576**, the second end portion **1594** of the extendable portion **1598** can include an aperture **1606**, the support bracket **1576** include a clevis **1608** with apertures **1609**, and a pin **1610** can be received in the apertures **1606** and **1609**.
(165) The extendable portion **1598** is moveable inwardly and outwardly relative to the base portion **1596**, and such inward and outward movement relative to the base portion **1596** serves to move the support bracket **1576** via movement of the two trucks **1574** along the track portion **1572** between at least a first position closer to the second end **1582** of the third upstanding portion **1516** to a second position closer to the first end **1580** of the third upstanding portion **1516**. As discussed below, the movement of the support bracket **1576** using the extension of the extendable portion **1598**

ultimately serves in adjusting the position of the patient's spine. Such adjustment can occur before, during, and/or after surgery using the surgical frame **1400**.

(166) The upper leg support **1402** also includes a support assembly **1620** that is carried by the support bracket **1576**. The support assembly **1620** includes a first support post **1622**, a second support post **1624**, and a connecting bracket **1626** connecting the first support post **1622** and the second support post **1624** to one another. The support assembly **1620** also includes a first support block **1630**, a second support block **1632**, a third support block **1634**, a fourth support block **1636**, a first support plate **1640**, a second support plate **1642**, and a third support plate **1644**. Each of the first support plate **1640**, the second support plate **1642**, and the third support plate **1644** include upper surfaces **1646**A, **1646**B, and **1646**C, respectively, that can be used to contact the upper legs of the patient. The upper surfaces **1646**A and **1646**C can be covered with padding (not shown) for contacting portions of the patient's upper legs, and the padding can include pressure sensors (not shown) incorporated therein. The resulting pressure sensing padding can be used to determine if undue stress is placed on the patient P via articulation of the upper leg support **1402**.

(167) The first support plate **1640** and the second support plate **1642**, as discussed below, are moveable with respect to the support bracket **1576** and the third support plate **1644**. As depicted in FIG. **46**, the third support plate **1644** is attached to the support bracket **1576**, and the second support plate **1642** is positioned such that it can move underneath the third support plate **1644**. As such, during movement of the first support plate **1640** and the second support plate **1642**, the area defined by the upper surfaces **1646**A, **1646**B, and **1646**C can be respectively decreased or increased as the second support plate **1642** is moved under or out from under the third support plate **1644**.

(168) As depicted in FIG. **45**, the first support plate **1640** is attached to the first support block **1630** and the second support block **1632**, and the first support block **1630** is moveable along the first support post **1622** and the second support block **1632** is moveable along the second support post **1624**. Furthermore, the second support plate **1642** is attached to the third support block **1634** and the fourth support block **1636**, and the third support block **1634** is moveable along the first support post **1622** and the fourth support block **1636** is moveable along the second support post **1624**.

(169) The first and second support blocks **1630** and **1632** include apertures **1650** and **1652** for receiving the first and second support posts **1622** and **1624**, respectively, and the third and fourth support blocks **1634** and **1636** include apertures **1654** and **1656** for receiving the first second support posts **1622** and **1624**, respectively. The first support plate **1640** and the second support plate **1642** are moveable inwardly and outwardly relative to the support bracket **1576** and the third support plate **644** via movement of the first and third support blocks **1630** and **1634** on the first support post **1622** and via movement of the second and fourth support blocks **1632** and **1636** on the second support post **1624**.

(170) The support assembly **1620** also includes a threaded shaft **1660** that is retained in position between the support bracket **1576** and the connecting bracket **1626**. As discussed below, the threaded shaft **1660** is used to constrain movement of the first support plate **1640** and the second support plate **1642** relative to the third support plate **644** and the main beam **1404**.

(171) Furthermore, the first support plate **1640** includes a first support collar **1664** opposite from the upper surface **1646**A, and the second support plate **1642** includes a second support collar **1666** opposite from the upper surface **1646**B. The first support collar **1664** includes a first aperture **1670** that can include threads complementary to those of the threaded shaft **1660**, and the second support collar **1666** includes a second aperture **1672** that can include threads complementary to those of the threaded shaft **1660**. The threaded shaft **1660** can be received in the first aperture **1670** and the second aperture **1672**. The first support collar **1664** and the second support collar **1666** can include one or more latches (not shown) that can be engaged and disengaged from the threaded shaft **1660** via actuation thereof. The one or more latches can be attached to the first support collar **1664** and/or the second support collar **1666**, and a user can actuate the one or more latches to engage or

disengage the threaded shaft **1660** to correspondingly prevent movement or allow movement of the first support collar **1664** and the second support collar **1666** along the threaded shaft **1660**. When the one or more latches are engaged, the interactions of the one or more latches with the threaded shaft **1660** prevent movement of the first support plate **1640** and the second support plate **1642** relative to the third support plate **1644**. When the one or more latches are disengaged, the first support plate **1640** and the second support plate **1642** can move relative to the third support plate **1644**. Rather than using the threaded shaft **1660**, a shaft with catches and/or teeth to which the one or more latches can be engaged and disengaged.
(172) Alternatively, a motor/transmission/actuator (not shown) can be used to facilitate rotation of the threaded shaft **1660**, and rotation of the threaded shaft **1660** and the interaction in the first aperture **1670** and the second aperture **1672** causes corresponding movement of the first support plate **1640** and the second support plate **1642**. As such, rotation of the threaded shaft **1660** via actuation of the motor/transmission/actuator is translated into movement of the first support plate **1640** and the second support plate **1642**. To illustrate, the threads of the threaded shaft **1660**, the first aperture **1670**, and the second aperture **1672** can be configured such that clockwise rotation of the threaded shaft **1660** via actuation of the motor/transmission/actuator causes inward movement of the first support plate **1640** and the second support plate **1642**, and counterclockwise rotation of the threaded shaft **1660** via actuation of the motor/transmission/actuator causes outward movement of the first support plate **1640** and the second support plate **1642**. The inward and outward movement of the first support plate **1640** and the second support plate **1642** is relative to the third support plate **1644** and the main beam **1404**.
(173) The movement of the first support plate **1640** and the second support plate **1642** ultimately serves in adjusting a total width of a combination of the first support plate **1640**, the second support plate **1642**, and the third support plate **1644**. Adjustment of the combined width of the first support plate **1640**, the second support plate **1642**, and the third support plate **1644** affords the accommodation of differently sized patients on the upper leg support **1402**.
(174) The movement of the componentry of the upper leg support **1402** can be effectuated via manual adjustment and/or controlled automation. To illustrate, the first arm portion **1450** includes the extendable portion **1472** that is moveable with respect to the base portion **1470** thereof, the telescoping shaft portion **1530** includes the extendable portion **1538** that is moveable with respect to the base portion **1536** thereof, the telescoping shaft portion **1590** includes the extendable portion **1598** that is moveable with respect to the base portion **1596** thereof, and the motor/transmission/actuator facilitates movement of the first support plate **1640** and the second support plate **1642** is relative to the third support plate **1644** and the main beam **1404**.
(175) Such reconfiguration of the upper leg support **1402** can be actuated using the manual adjustment and/or the controlled automation, and as discussed below, the extension and retraction of the extendable portion **1472**, the extendable portion **1538**, and the extendable portion **1598** via such actuation can be used to both adjust the adjustable center of rotation CR to accommodate patients having different body sizes, and to facilitate flexing of the lumbar spine to afford surgical access thereto via manipulation of portions thereof. The extension/retraction of the extendable portion **1472** serves to change the angle of the first arm portion **1450** and the second arm portion **1452** relative to one another, the extension/retraction of the extendable portion **1538** serves to change the angle of the platform portion **1454** relative to the second arm portion **1452**, and the extendable portion **1598** serves to change position of the bracket **1576** (which supports the first support plate **1640**, the second support plate **1642**, and the third support plate **1644**) relative to the platform portion **1454**.
(176) Using the upper leg support **1402**, the position of the patient's upper legs can be altered, which correspondingly affects the flexure of the lumbar spine of the patient P, and care should be taken to prevent unwanted torsion thereof when manipulating the patient's spine. To illustrate, the amounts of extension/retraction of the extendable portion **1472**, the extendable portion **1538**, and

the extendable portion **1598** can be constrained with respect to one another to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support **1402**. As such, the amounts of extension/retraction of the extendable portion **1472**, the extendable portion **1538**, and the extendable portion **1598** can be contingent upon one another to facilitate such approximate preservation.
(177) A controller (not shown) with a user interface (not shown) can be used to control the constrained/contingent extension and/or retraction of the extendable portion **1472**, the extendable portion **1538**, and the extendable portion **1598** via the controlled automation. Such extension and/or retraction, as depicted in FIGS. **48** and **49**, affords positioning and repositioning of the support assembly **1620**. Furthermore, because patients' heights can vary, the amounts of extension/retraction of the extendable portion **1472**, the extendable portion **1538**, and the extendable portion **1598** can be altered to accommodate these different heights while still being constrained/contingent upon one another to provide for the desired amount of manipulation of portions of the lumbar spine during articulation of the upper leg support **1402**.
(178) The controller with input via the user interface can allow the user to select the desired center of rotation and the desired amount of manipulation of the portions of the lumbar spine. To illustrate, the user interface can be used to display a graphical or actual representation of the patient's spine, and the user interface can permit the user to input the desired center of rotation and the desired amount of manipulation by, for example, highlighting a portion of the graphical or actual representation of the patient's spine on the user interface. The selection of the desired amount of manipulation can allow the user to select where the forces applied via the actuation of the extendable portion **1472**, the extendable portion **1538**, and the extendable portion **1598** are applied during flexure of the patient's spine. In addition to or alternatively to use of the user interface, a navigation tool interconnected with the controller and/or the user interface can be positioned on or adjacent the patient's spine to facilitate inputting of the desired center of rotation and the desired amount of manipulation. The inputting of the desired center of rotation and the desired amount of manipulation can be done with the main beam **404** and the patient P supported on the main beam **1404** in various rotational positions including, but not limited to, prone, lateral, and supine positions.
(179) When the upper leg support **1402** is articulated such that the lumbar spine of the patient P is in an unflexed neutral position, as depicted in FIG. **48**, the controller can be used to extend/retract the extendable portion **1472** (of the first arm portion **1450**), the extendable portion **1538** (of the telescoping shaft portion **1540**), and the extendable portion **1598** (of the telescoping shaft portion **1590**) such that the first arm portion **1450**, the telescoping shaft portion **1540**, and the telescoping shaft portion **1590** have lengths that accommodate the height of the patient P. Furthermore, when the upper leg support **1402** is articulated such that the lumbar spine of the patient P has a 30 degree flex, as depicted in FIG. **49**, the controller can be used to extend/retract the extendable portion **1472** (of the first arm portion **1450**), the extendable portion **1538** (of the telescoping shaft portion **1530**), and the extendable portion **1598** (of the telescoping shaft portion **1590**) such that the first arm portion **1450**, the telescoping shaft portion **1540**, and the telescoping shaft portion **1590** have lengths that flex the lumbar spine of the patient P to afford surgical access thereto. Moreover, during the transition between the positions of FIGS. **48** and **49**, the controller can serve to prevent unwanted torsion of the lumbar spine during articulation of the upper leg support **1402** by properly adjusting the amounts of extension/retraction of the extendable portion **1472**, the extendable portion **1538**, and the extendable portion **1598**.
(180) The relative amounts of extension/retraction can be provided for different patient heights and different degrees of flex of the lumbar spine, and can be included as presets in the controller. Thus, using the user interface of the controller, the operator of the surgical frame **1400** can input the height of the patient P, and input the desired degree of flexure of the lumbar spine, and the controller can actuate the first arm portion **1450** (to extend or retract the extendable portion **1472**),

the telescoping shaft portion **1530** (to extend or retract the extendable portion **1538**), and the telescoping shaft portion **1590** (to extend or retract the extendable portion **1598**) the appropriate amounts to provide such flexion, while also preventing unwanted torsion of the patient's spine. As discussed above, the extension/retraction of the extendable portion **1472** serves to change the angle of the first arm portion **1450** and the second arm portion **1452** relative to one another, the extension/retraction of the extendable portion **1538** serves to change the angle of the platform portion **1454** relative to the second arm portion **1452**, and the extendable portion **1598** serves to change position of the bracket **1576** (which supports the first support plate **1640**, the second support plate **1642**, and the third support plate **1644**) relative to the platform portion **1454**. During such manipulation of the patient's spine using the upper leg support **1402**, the lengths of the first arm portion **1450**, the telescoping shaft portion **1530**, and the telescoping shaft portion **1590** may each be alternatingly increased/decreased or decreased/increased to provide for the desired adjustable center of rotation CR. As such, the operator can use the controller to manipulate the upper leg support **1402** to flex the lumbar spine of the patient P into position for surgery, while simultaneously inhibiting the unwanted torsion of the patient's spine caused be reconfiguration of the upper leg support **1402**.

(181) In addition to the upper leg support **1402**, the surgical frame **1400**, as depicted in FIGS. **42** and **47** includes the lower leg support **1416** that supports the lower legs of the patient P. The lower leg support includes a support plate portion **1680**, a first arm portion **1682**, a second arm portion **1684**, a first plate portion **1686**, a second plate portion **1688**, and a connecting rib **1690**. The first arm portion **1682** and the second arm portion **1684** include first end portions **1692** and second end portions (not shown). The first end portions **1692** can include apertures **1696** facilitating attachment thereof to the third portion **1410** via receipt of the pin **1492** therein. Furthermore, the second end portions can be attached to the first plate portion **1686**, and the first plate portion **1686** can be attached to the first arm portion **1450**. As such, portions of the lower leg support **1416** can move with the articulation of the first arm portion **1450**. The first plate portion **1686** connects the second plate portion **1688** and the connecting rib **1690** to one another, and the connecting rib **1690** attaches the support plate portion **1680** to the first plate portion **1686**. As depicted in FIG. **42**, the connecting rib **1690** spaces the support plate portion **1680** from the first plate portion **1686**. The support plate portion **1680** can be used to support the patient's lower legs thereon.

(182) Further, other types of mechanism or actuators, such as servomotors, can be used or configuration to provide for the mechanical articulations and movements necessary to support the biomechanical manipulations of the spine described herein.

(183) It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and the accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes of methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspect of this disclosure are described as being performed by a single module or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or modules associated with, for example, a medical device.

## Claims

1. A method of adjusting a position of a patient supported on a surgical frame, the method comprising: positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate portion; extending a first arm portion relative to a second arm portion to adjust a position of a platform portion supporting the support plate portion relative to a portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the

surgical frame and a second end attached relative to the platform portion, and the second arm portion including a first end attached relative to the platform portion and a second end attached relative to the portion of the surgical frame; extending a telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby relative to at least one of the first arm portion and the second arm portion, the telescoping shaft including a first end attached relative to the second arm portion and a second end attached relative to the platform portion; and adjusting a position of a lumbar portion of a spine of the patient by coordinating extension of the first arm portion and the telescoping shaft to adjust the position of the platform and the support plate portion supported thereby downwardly and rotatably relative to the portion of the surgical frame to correspondingly adjust the upper legs of the patient supported by the support plate portion.

2. The method of claim 1, wherein the second end of the first arm portion and the first end of the second arm portion are pivotally attached to one another and to the platform portion.

3. The method of claim 2, wherein during extension of the first arm portion, the first arm portion and the second arm portion are pivoted with respect to one another.

4. The method of claim 3, wherein during extension of the telescoping shaft, the platform portion is rotated relative to the at least one of the first arm portion and the second arm portion.

5. The method of claim 1, wherein the first arm portion and the second arm portion are pivoted with respect to the portion of the surgical frame and one another during extension of the first arm portion, and the platform portion is rotated relative to the at least one of the first arm portion and the second arm portion during extension of the telescoping shaft.

6. The method of claim 1, further comprising rotating the portion of the surgical frame to position and reposition the portion of the surgical frame relative to remaining portions of the surgical frame.

7. The method of claim 1, wherein the support plate portion is attached to the platform portion via a support bracket, and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion, and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.

8. A method of adjusting a position of a patient supported on a surgical frame, the method comprising: supporting upper legs of the patient on a support plate portion of the surgical frame; supporting the support plate portion relative to a portion of the surgical frame by a platform portion, a first arm portion, and a second arm portion, the first arm portion and the second portion being pivotal with respect to one another via pivotal attachment thereof to the platform portion; extending the first arm portion to adjust a position of the platform portion and the support plate portion supported thereby relative to the portion of the surgical frame, the first arm portion including a first end attached relative to the portion of the surgical frame and a second end attached to the platform portion, and the second arm portion including a first end attached to the platform portion and a second end attached relative to the portion of the surgical frame; extending a telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby relative to the first arm portion and the second arm portion, the telescoping shaft including a first end attached to second arm portion and a second end attached to the platform portion; and adjusting a position of a lumbar portion of a spine of the patient by coordinating extension of the first arm portion and the telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby downwardly and rotably relative to the portion of the surgical frame to correspondingly adjust the upper legs of the patient supported by the support plate portion.

9. The method of claim 8, wherein during extension of the first arm portion, the first arm portion and the second arm portion are pivoted with respect to one another.

10. The method of claim 9, further comprising retracting the first arm portion to further adjust the position of the platform portion relative to the surgical frame.

11. The method of claim 8, wherein during extension of the telescoping shaft, a rotational position of the platform portion is adjusted.
12. The method of claim 8, further comprising rotating the portion of the surgical frame to position and reposition the portion of the surgical frame relative to remaining portions of the surgical frame.
13. The method of claim 8, wherein the support plate portion is attached to the platform portion via a support bracket; and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion; and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.
14. A method of adjusting a position of a patient supported on a surgical frame, the method comprising: pivotally attaching each of a first arm portion and a second arm portion to a portion of the surgical frame and to a platform portion; positioning the patient on the surgical frame by supporting upper legs of the patient on a support plate portion supported by the platform portion; pivoting and extending the first arm portion relative to the surgical frame and pivoting the second arm portion relative to the surgical frame to adjust the position of the platform portion relative to the portion of the surgical frame; and extending a telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby relative to at least one of the first arm portion and the second arm portion, the telescoping shaft including a first end attached relative to the second arm portion and a second end attached relative to the portion of the platform portion; and adjusting a position of a lumbar portion of a spine of the patient by coordinating extension of the first arm portion and the telescoping shaft to adjust the position of the platform portion and the support plate portion supported thereby downwardly and rotatably relative to the portion of the surgical frame to correspondingly adjust the upper legs of the patient supported by the support plate portion.
15. The method of claim 14, wherein the pivoting of the first arm portion is afforded by extension of the first arm portion.
16. The method of claim 14, wherein during extension of the telescoping shaft, the platform portion is rotated relative to the first arm portion.
17. The method of claim 14, wherein a first end the first arm portion is pivotally attached to the portion of the surgical frame and a second end of the first arm portion is attached to the platform portion, and a first end of the second arm portion is pivotally attached to the portion of the surgical frame and a second end of the second arm portion is attached to the platform portion.
18. The method of claim 17, wherein the support plate portion is attached to the platform portion via a support bracket; and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion, and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.
19. The method of claim 14, further comprising rotating the portion of the surgical frame to position and reposition the portion of the surgical frame relative to remaining portions of the surgical frame.
20. The method of claim 14, wherein the support plate portion is attached to the platform portion via a support bracket; and further comprising moving the support bracket along portions of the platform portion, and moving the support plate portion inwardly and outwardly relative to the support bracket, the support bracket being moveable along a track attached to the platform portion, and the support plate portion being moveable along at least one support post extending outwardly from the support bracket.